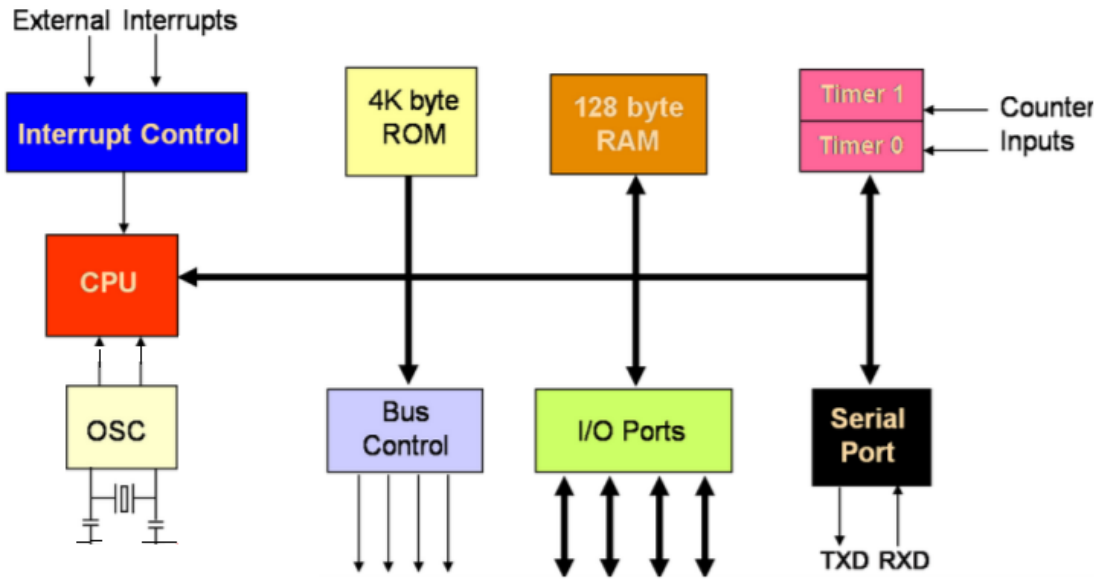
Exercise:

Timer/Counter in Stm32f407

Timer and Counter

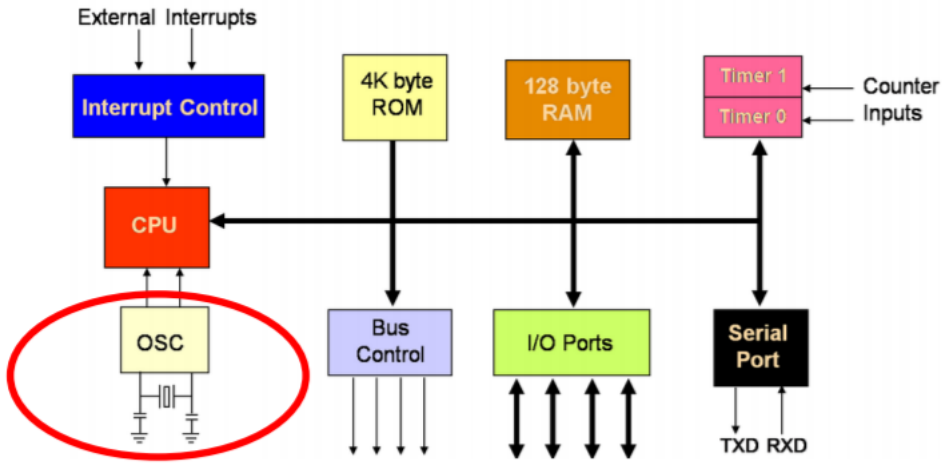
- **Overview:**

Microcontroller



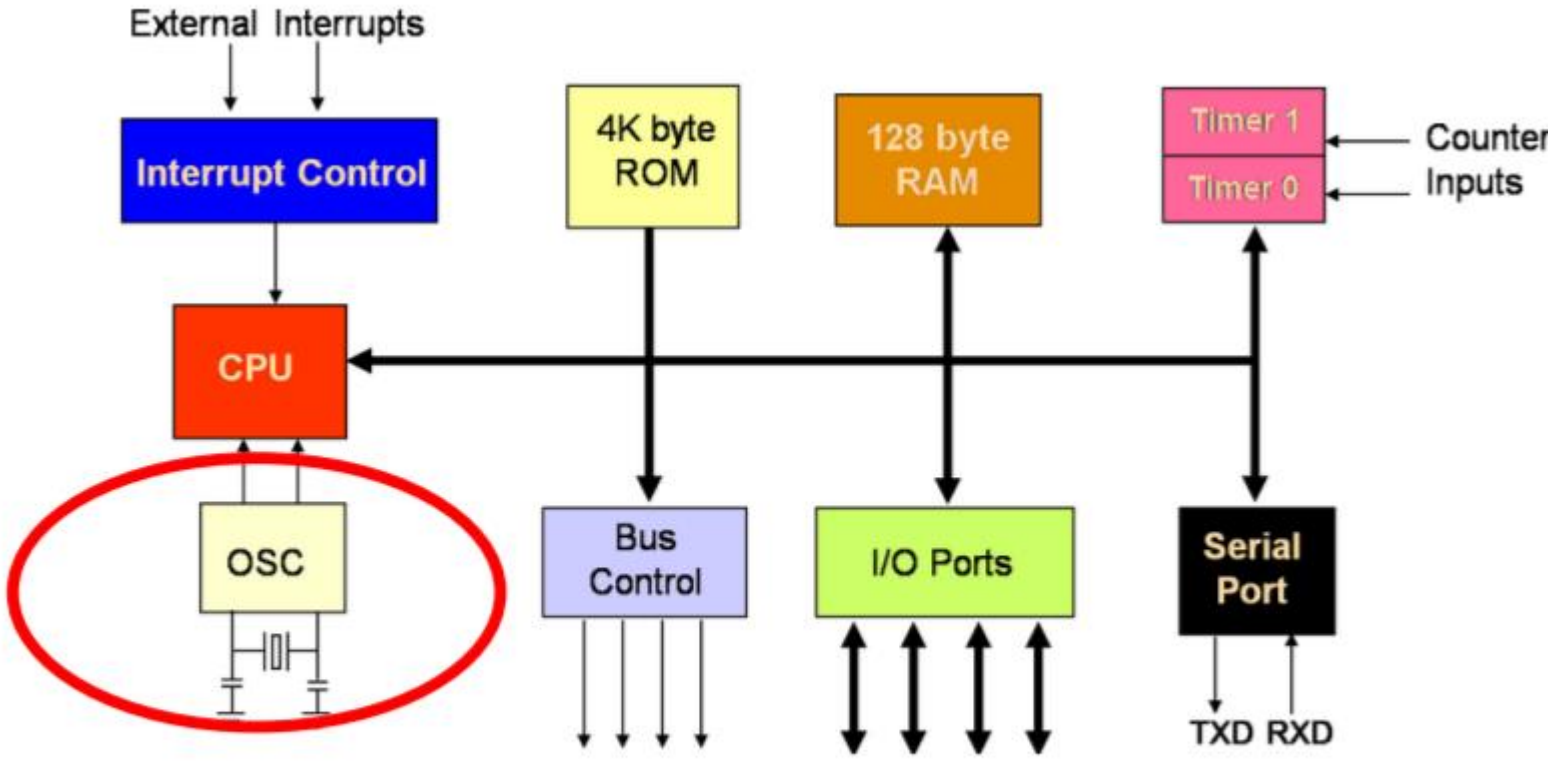
Timer and Counter

Oscillator



Oscillator for the Microcontroller is same as the Heart for Human beings

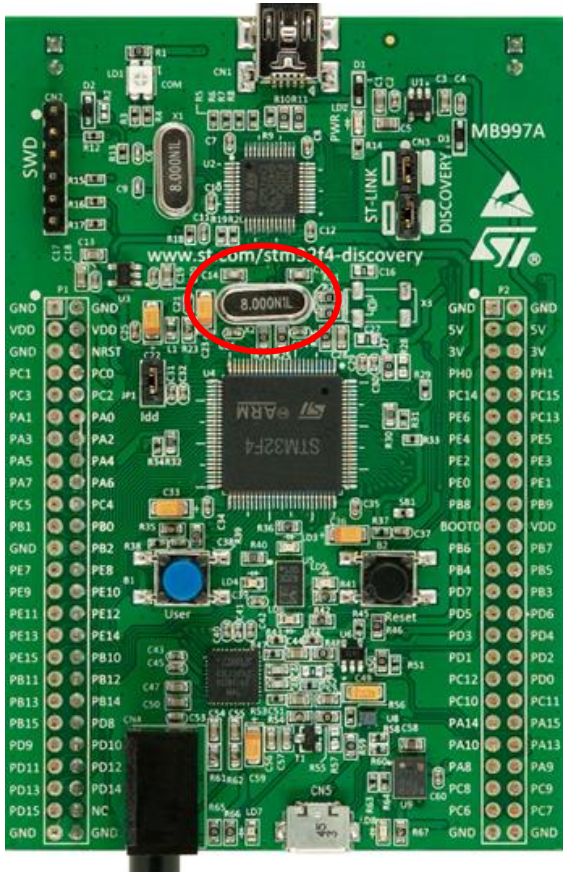
Timer and Counter



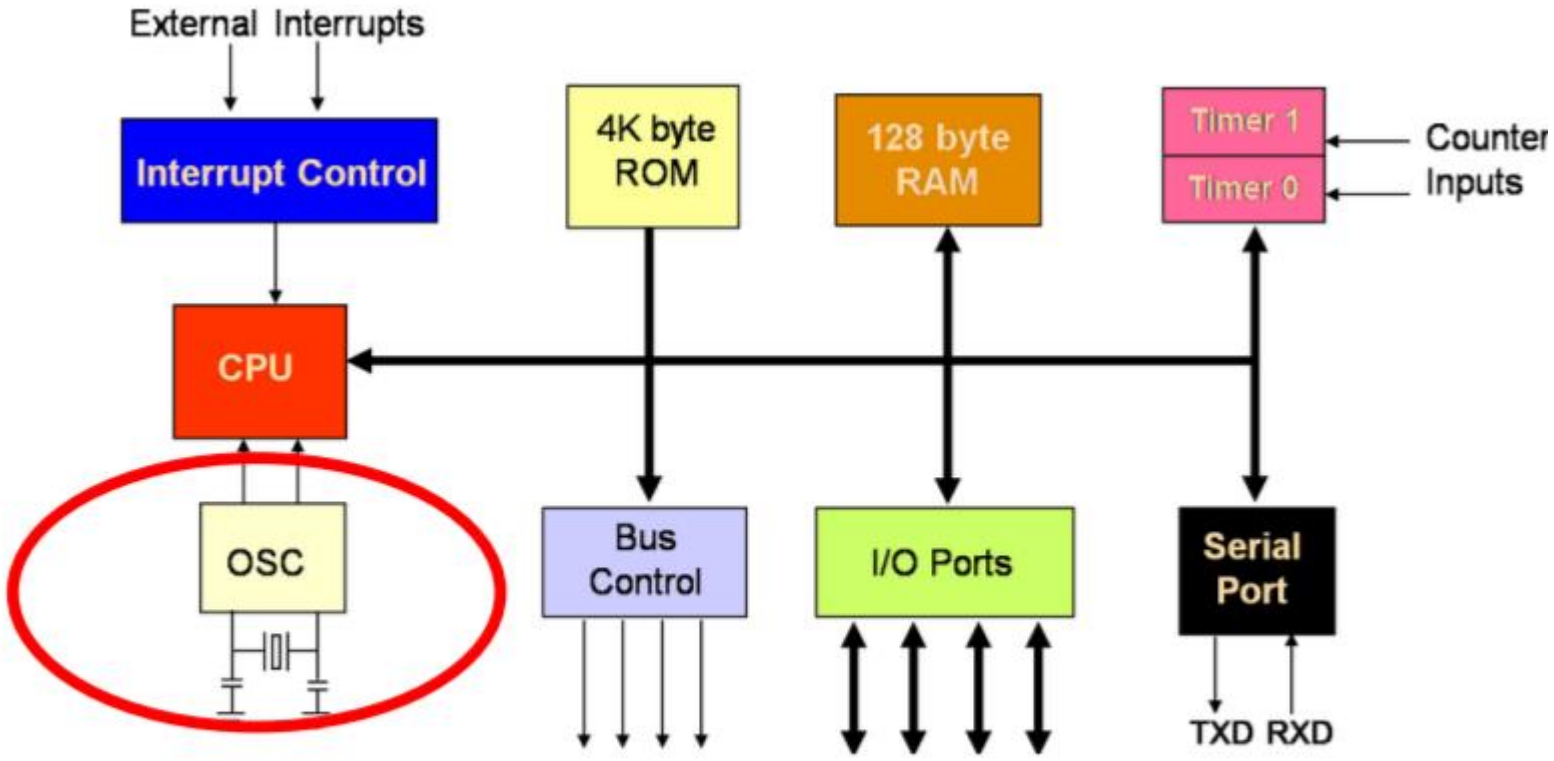
Timer and Counter

Purpose of an Oscillator:

- In general an oscillator is used to generate a clock signal.
- The clock signal generally changes between two or more states with the transition between the two states being very short (this transition is called an "edge").
- The purpose of the clock is to help all the parts of the microcontroller work together.
- The purpose of the clock is to make it easier to know when different parts of the microcontroller are going to change state.



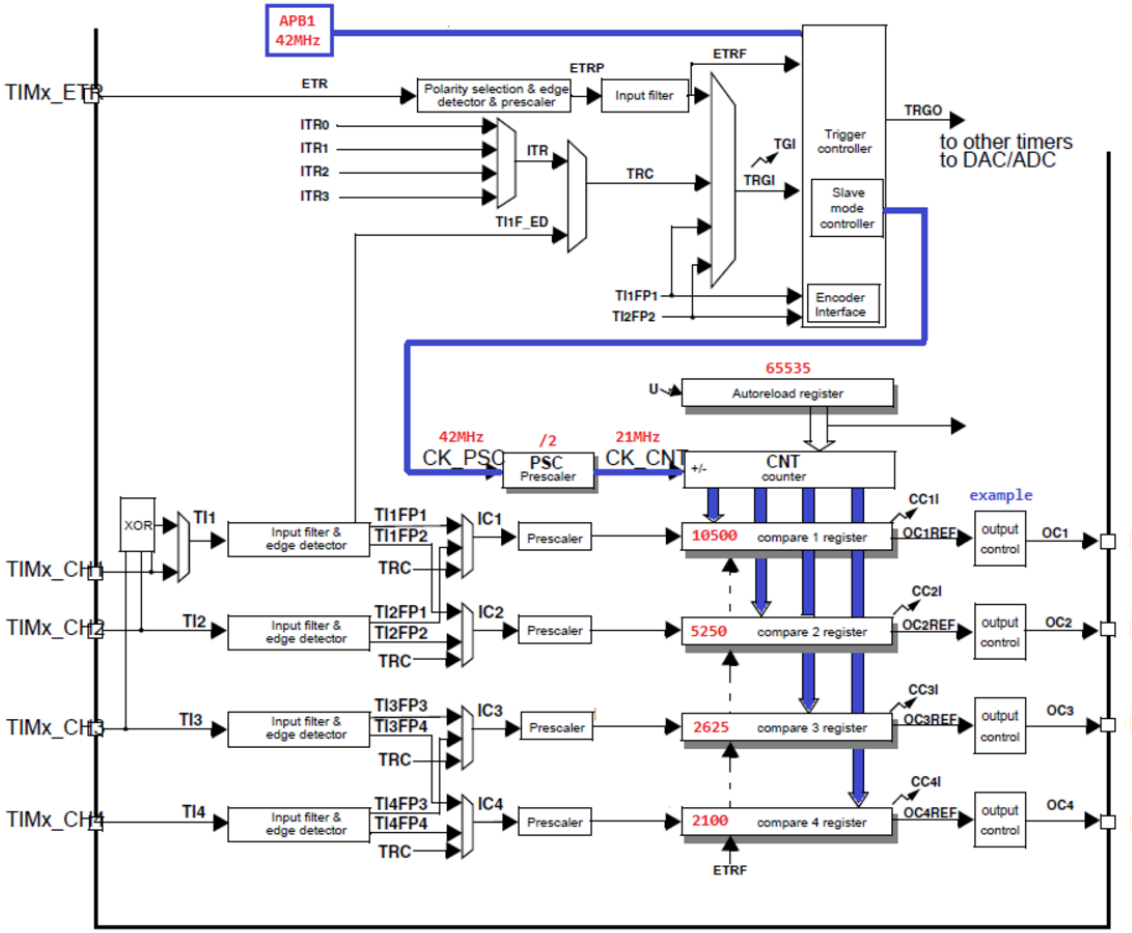
Timer and Counter

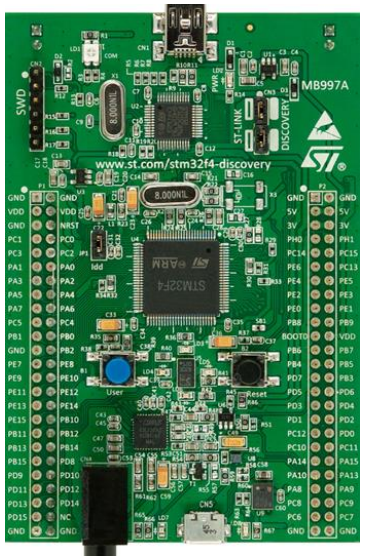


Timer and Counter

How does the structure of the timer looks like on the controller:

General-purpose timer block diagram





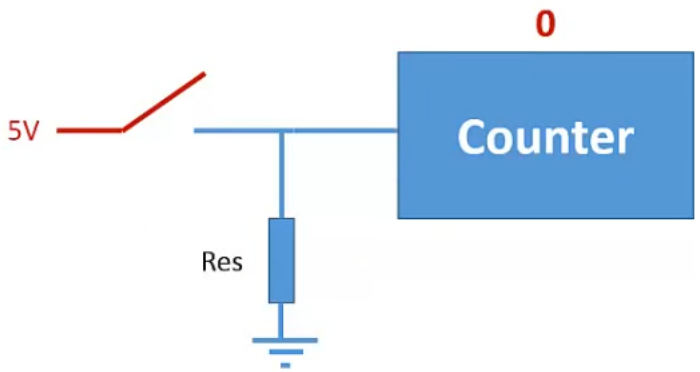
Timer and Counter

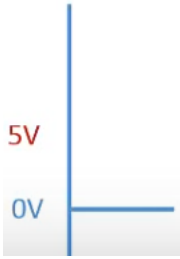
How does the structure of the timer looks like:

- **Timer:**

Timer and Counter

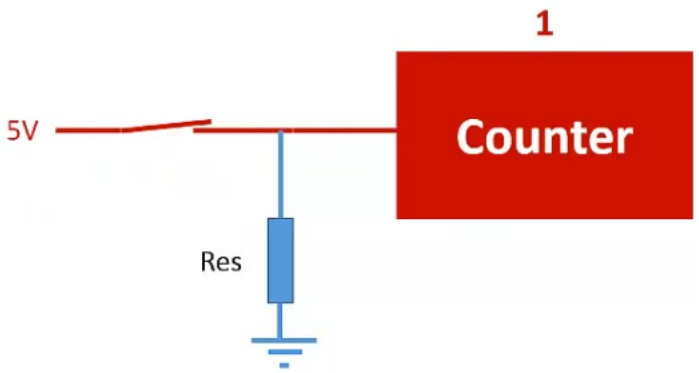
- **Counter:**

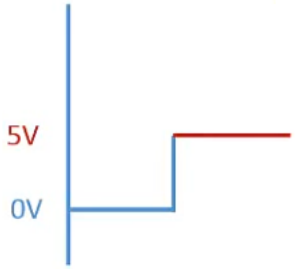




Timer and Counter

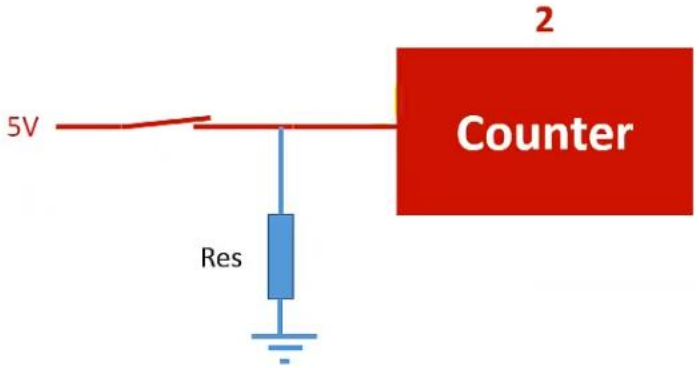
- **Counter:**

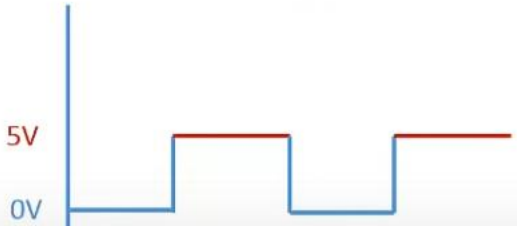




Timer and Counter

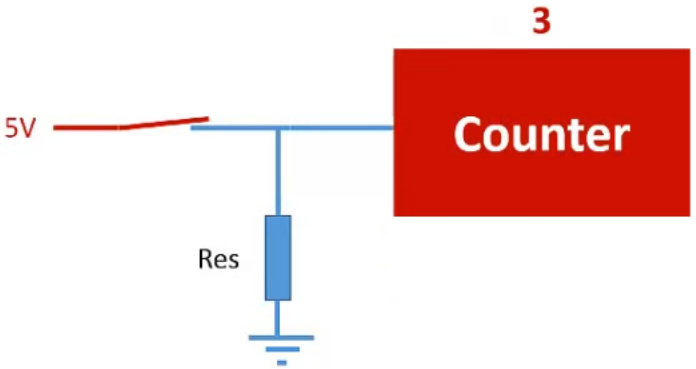
- **Counter:**

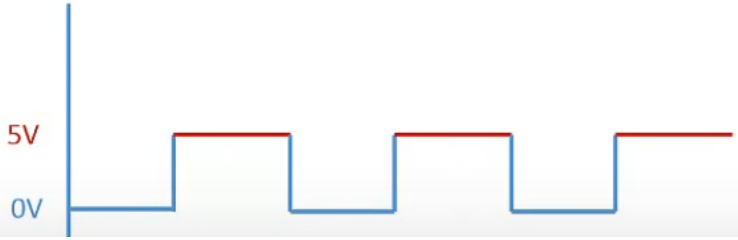




Timer and Counter

- **Counter:**





Timer and Counter

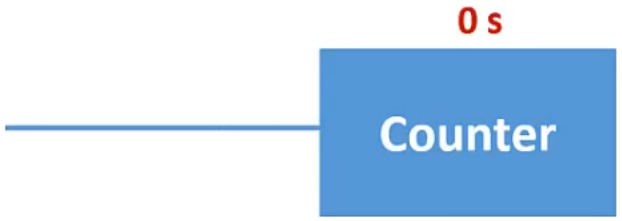
- **Counter:**

Counter

2 bit Counter ------> From 0 to 3 (2^2-1).
8 bit Counter ------> From 0 to 255 (2^8-1).
16 bit Counter ------> From 0 to 65535 (2^{16}-1).
32 bit Counter ------> From 0 to 4294967295 (2^{32}-1).
64 bit Counter ------> From 0 to 18446744073709551615 (2^{64}-1).

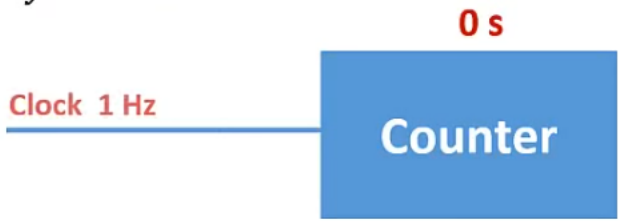
Timer and Counter

- **Counter :**



Timer and Counter

- **Counter :** $time = \frac{1}{frequency} = \frac{1}{1} = 1\ second$



Timer and Counter

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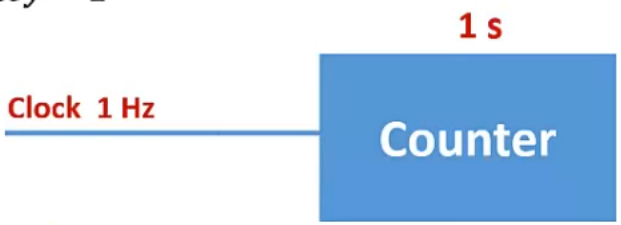

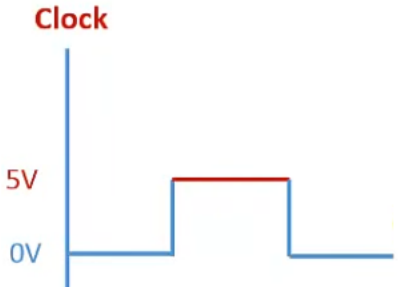

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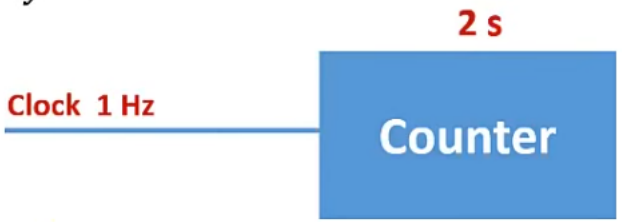

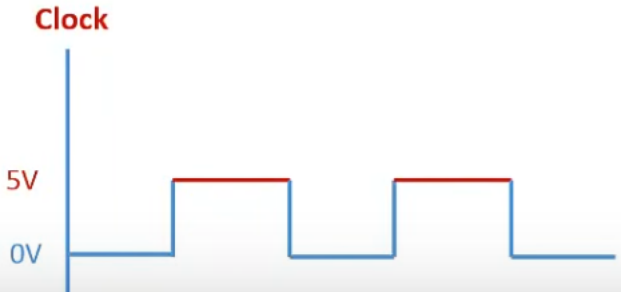

Timer and Counter

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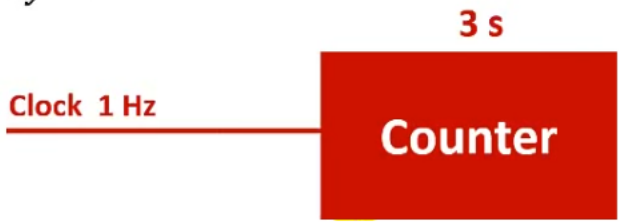
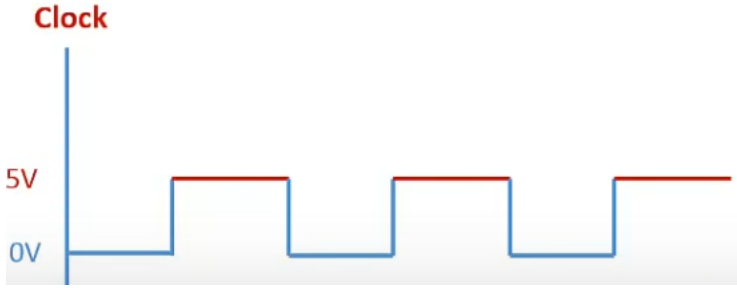




Timer and Counter

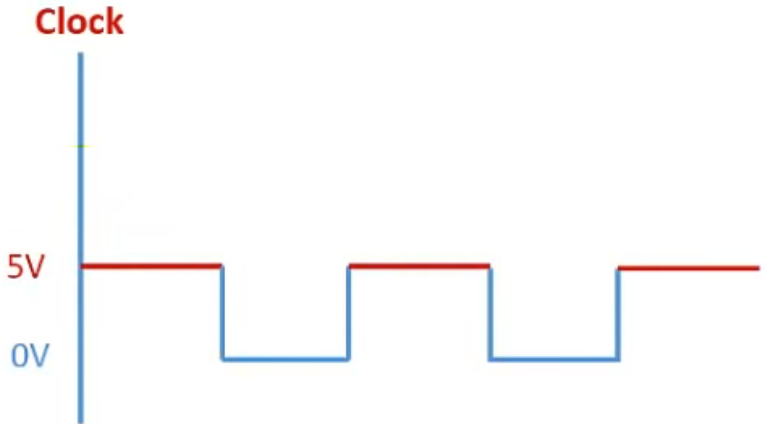
- **Timer:**

Timer and Counter

- **Timer:**

Timer and Counter

- **Timer:** $time = \frac{1}{frequency} = \frac{1}{16\ MHz} = 62.5$ ns



Timer and Counter

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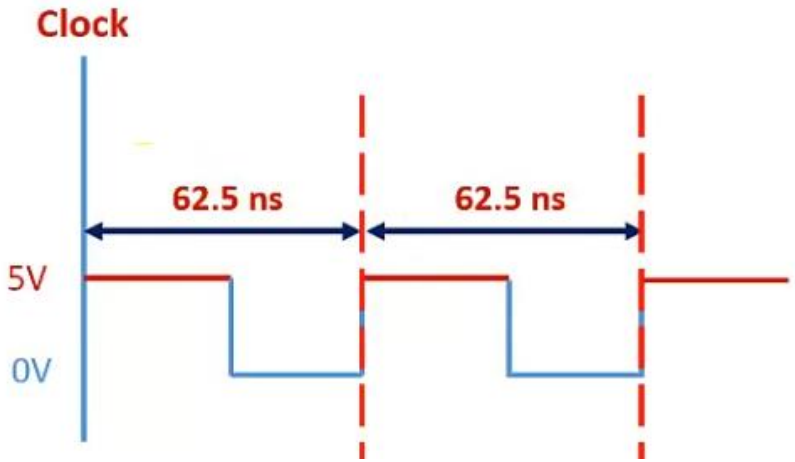

Timer and Counter

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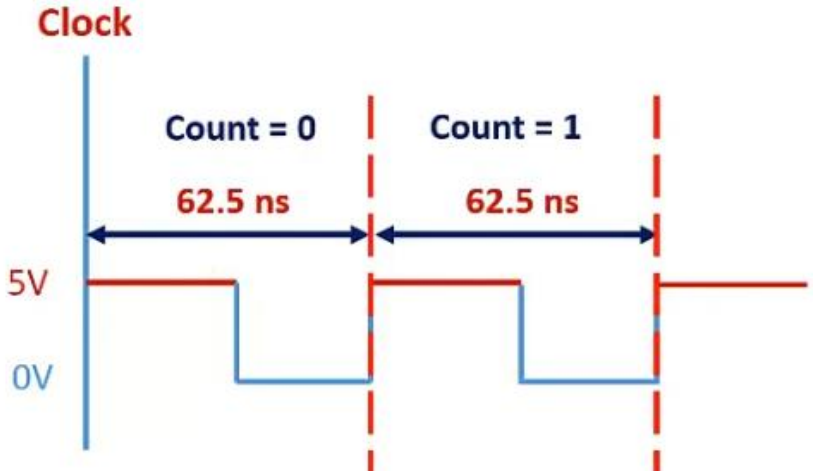
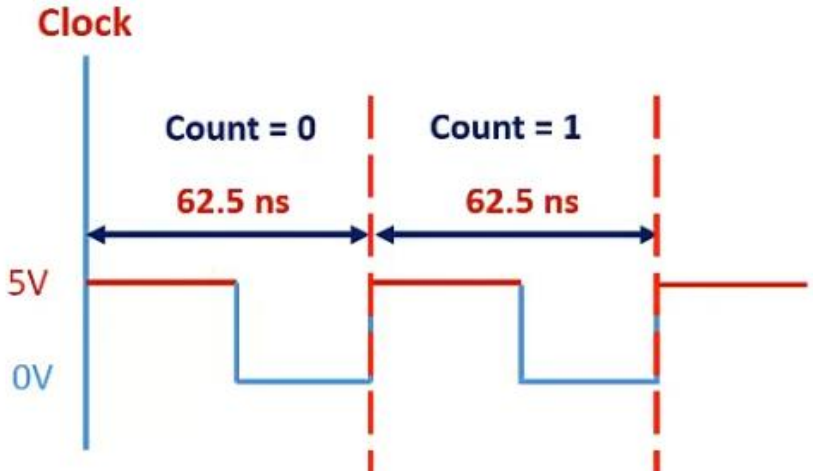


0.0000000625 Second ---> 1 Count

Timer and Counter

- **Timer:** $time = \frac{1}{frequency} = \frac{1}{16\ MHz} = 62.5$ ns



0.0000000625 Second ---> 1 Count

1 Second ---> 16000000 Count

Timer and Counter

- **Timer:**
  - **Problem:**

16 MHz

Atmega32

8 bit - 0 to 255

16 bit - 0 to 65535

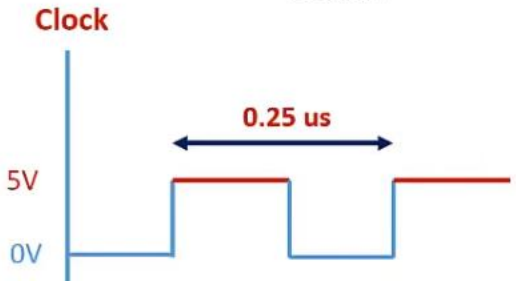
$$\frac{1}{16\ MHz} = 62.5\ \text{ns}$$

1 Second ---> 16000000 Count

Timer and Counter

- **Timer:**
  - **Solution 1 (Lower the clock frequency):**

$$\frac{1}{4\ MHz} = 0.25\ \text{us}$$



Timer and Counter

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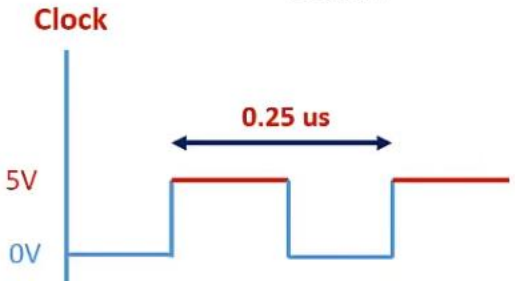
  - **Solution 1 (Lower the clock frequency):**

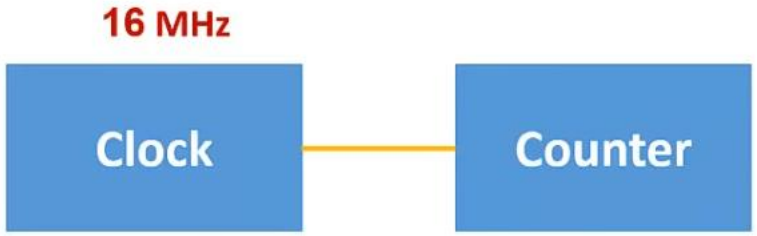
$$\frac{1}{4\ MHz} = 0.25\ \text{us}$$



1 Second ---> 4000000 Count

Timer and Counter

- **Timer:**
  - **Solution 2 (Use 32bit or 64bit Counter):**



TivaC TM4C123

16 bit - 0 to 65535
32 bit - 0 to 4294967295
64 bit - 0 to 18446744073709551615

$$\frac{1}{16\ MHz} = 62.5\ \text{ns}$$

1 Second ---> 16000000 Count

Timer and Counter

- **Timer:**
  - **Solution 3 (Use Prescaler):**

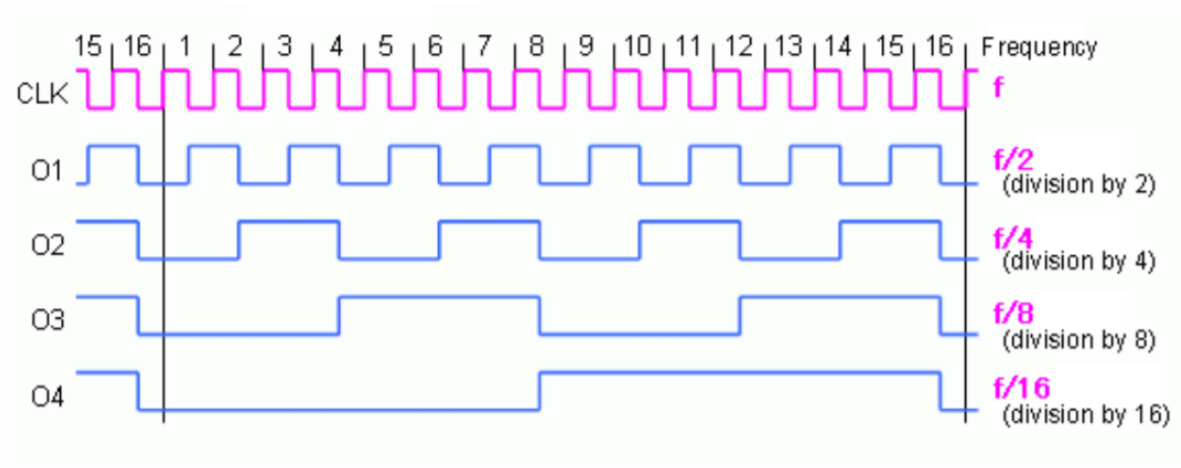
- **Timer:**
  - **Solution 3 (Use Prescaler):**

Factor

16 MHz

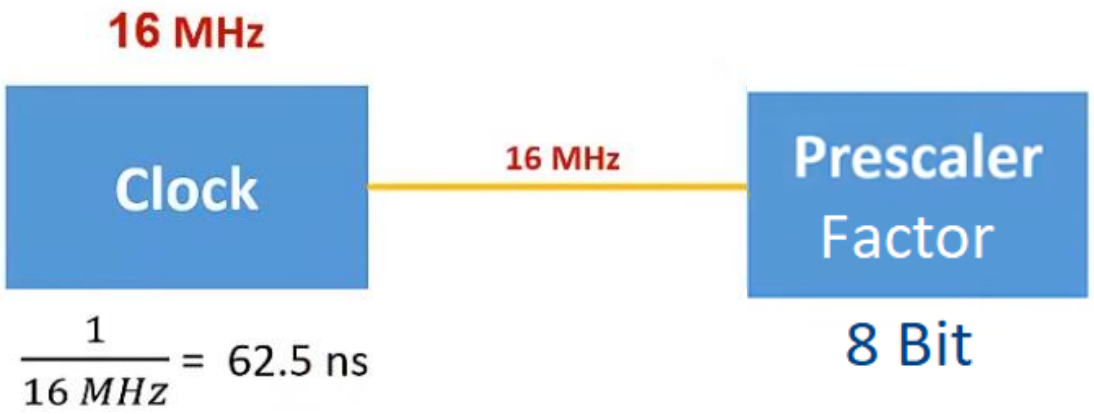
Clock

$$\frac{1}{16\ MHz} = 62.5\ \text{ns}$$



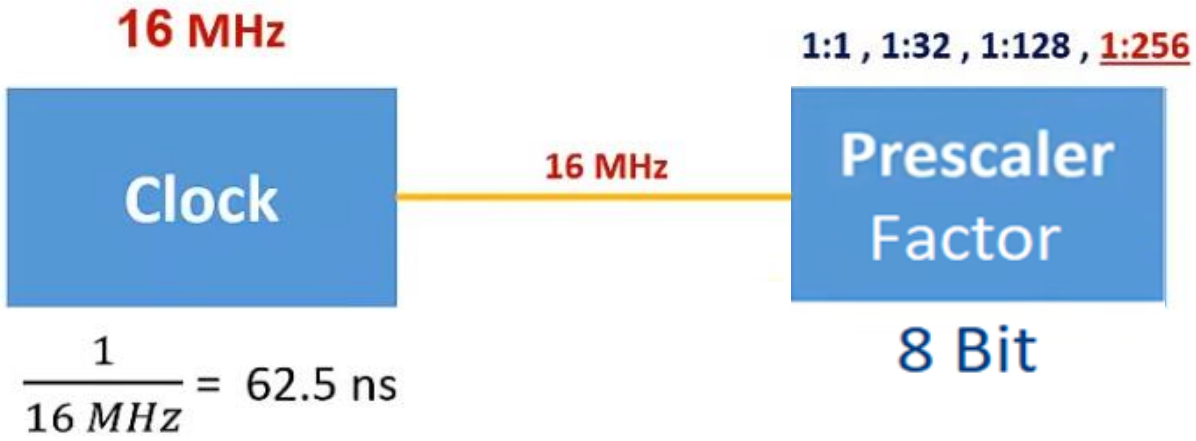
Timer and Counter

- **Timer:**
  - **Solution 3 (Use Prescaler):**



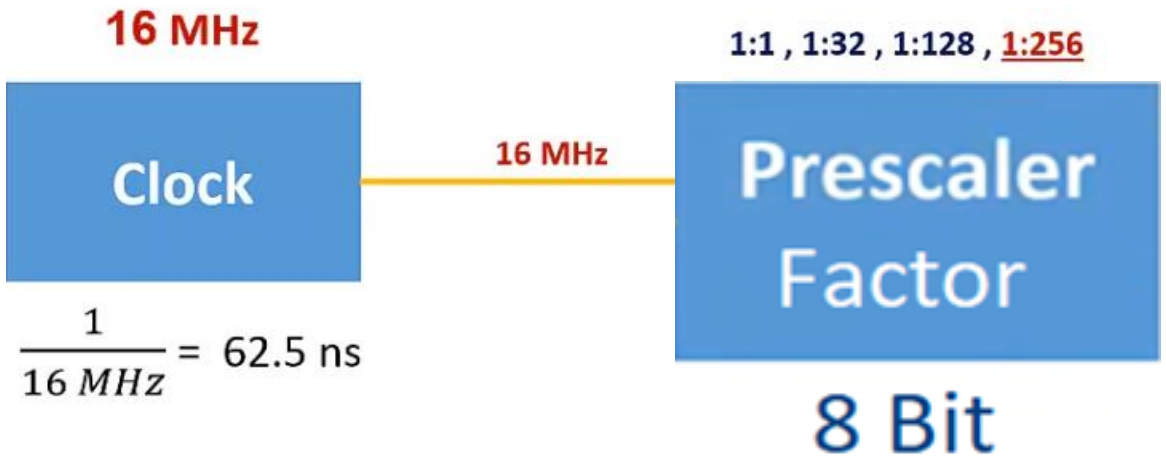
Timer and Counter

- **Timer:**
  - **Solution 3 (Use Prescaler):**



Timer and Counter

- **Timer:**
  - **Solution 3 (Use Prescaler):**

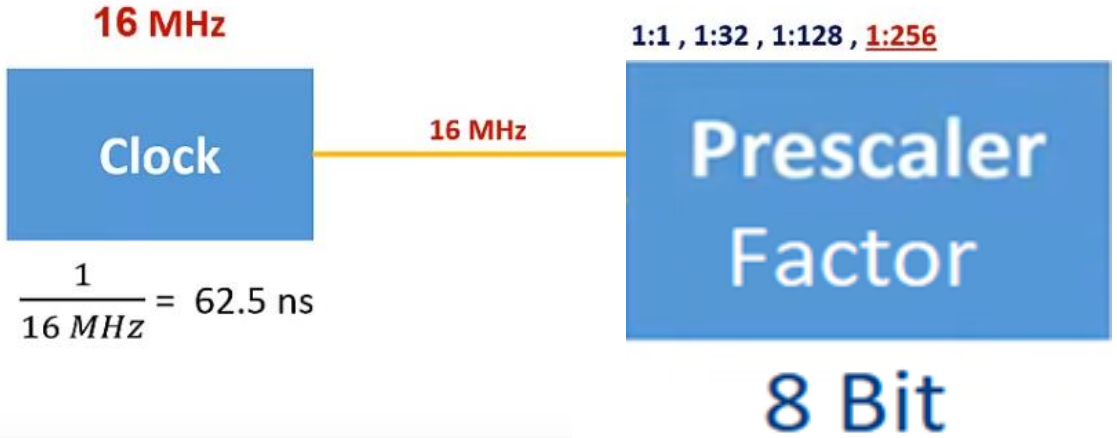


$$\frac{1}{16\ MHz} = 62.5\ \text{ns}$$

$$\frac{16\ MHz}{256} = 0.0625\ \text{MHz}$$

Timer and Counter

- **Timer:**
  - **Solution 3 (Use Prescaler):**



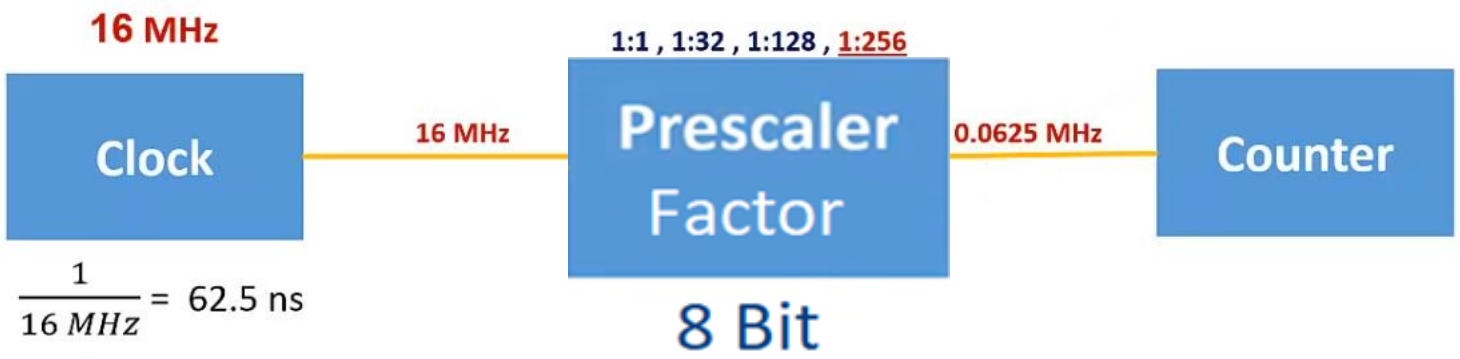
$$\frac{1}{16\ MHz} = 62.5\ \text{ns}$$

$$\frac{16\ MHz}{256} = 0.0625\ \text{MHz}$$

$$\frac{1}{0.0625\ MHz} = 16\ \text{us}$$

Timer and Counter

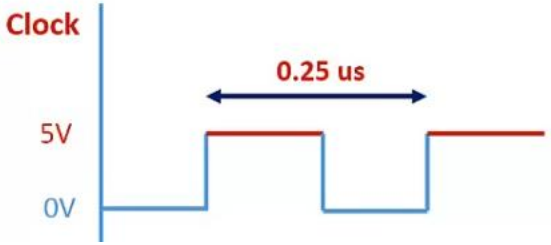
- **Timer:**
  - **Solution 3 (Use Prescaler):**



$$\frac{1}{16\ MHz} = 62.5\ \text{ns}$$

$$\frac{16\ MHz}{256} = 0.0625\ \text{MHz}$$

$$\frac{1}{0.0625\ MHz} = 16\ \text{us}$$



0.000016 S ---> 1 Count

Timer and Counter

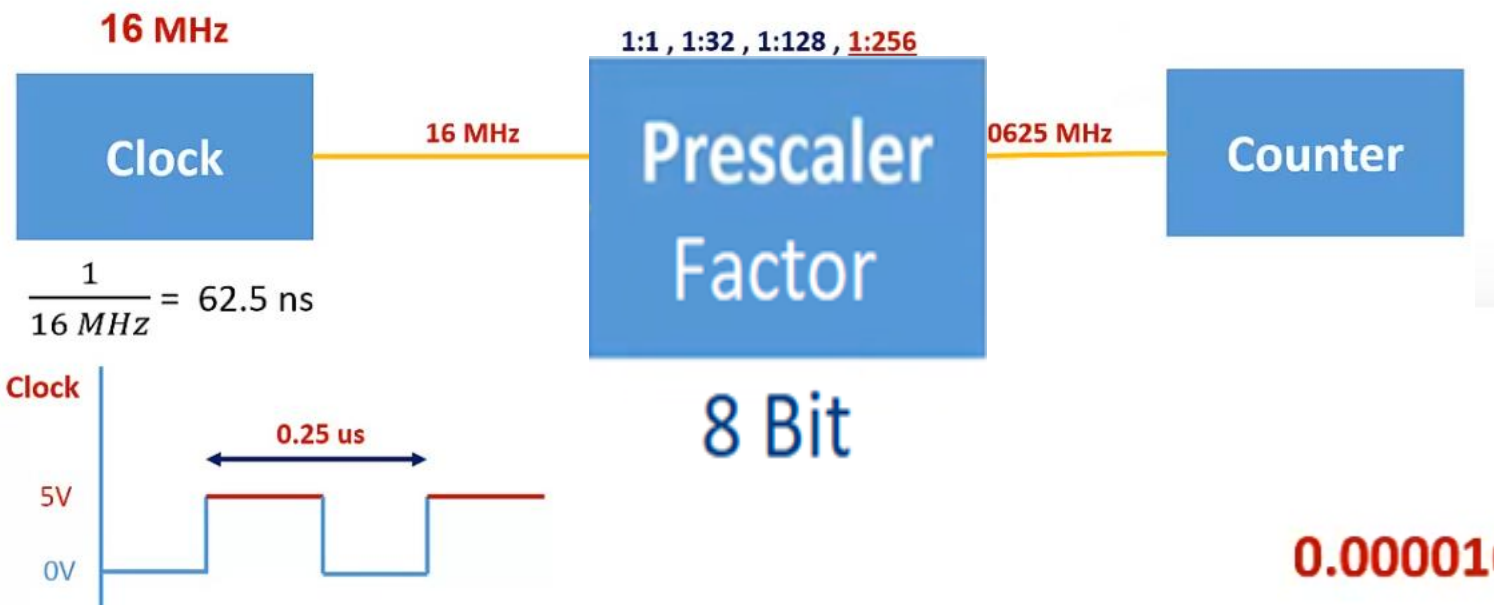
TivaC TM4C123

16 bit - 0 to 65535

32 bit - 0 to 4294967295

64 bit - 0 to 18446744073709551615

- **Timer:**
  - **Solution 3 (Use Prescaler):**



$$\frac{16\ MHz}{256} = 0.0625\ \text{MHz}$$

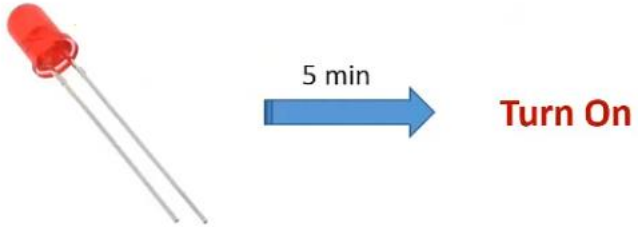
$$\frac{1}{0.0625\ MHz} = 16\ \text{us}$$

0.000016 S ---> 1 Count

1 S ---> 62500 Count

Timer and Counter

- **Timer:**
  - **Timer Usage:**
    1. **Delay.**
    2. **Periodic Task.**
    3. **Real Time Clock.**
    4. **Measure Signals Length.**
    5. **PWM.**
    6. **Etc..**



Timer and Counter

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  - **Timer Usage:**
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    2. Periodic Task.
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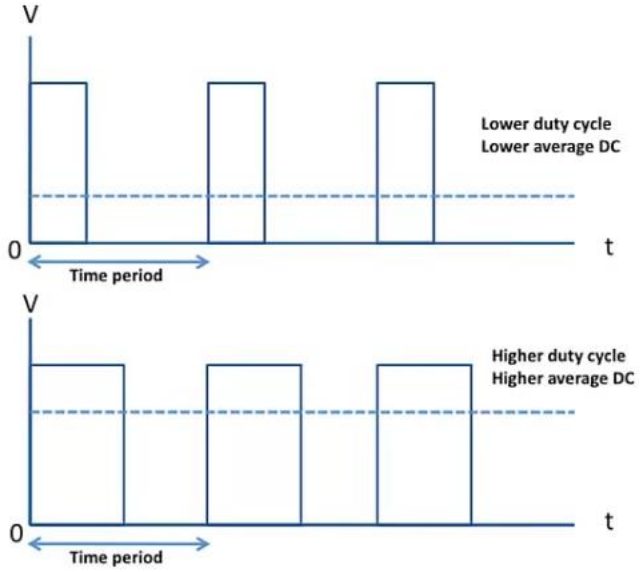

Timer and Counter

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  1. Delay.
  2. Periodic Task.
  3. Real Time Clock.
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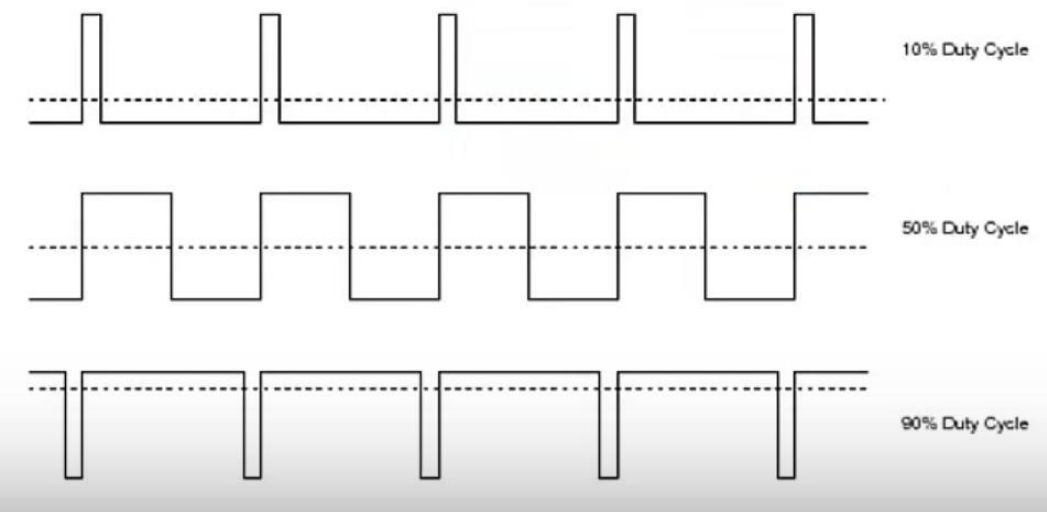
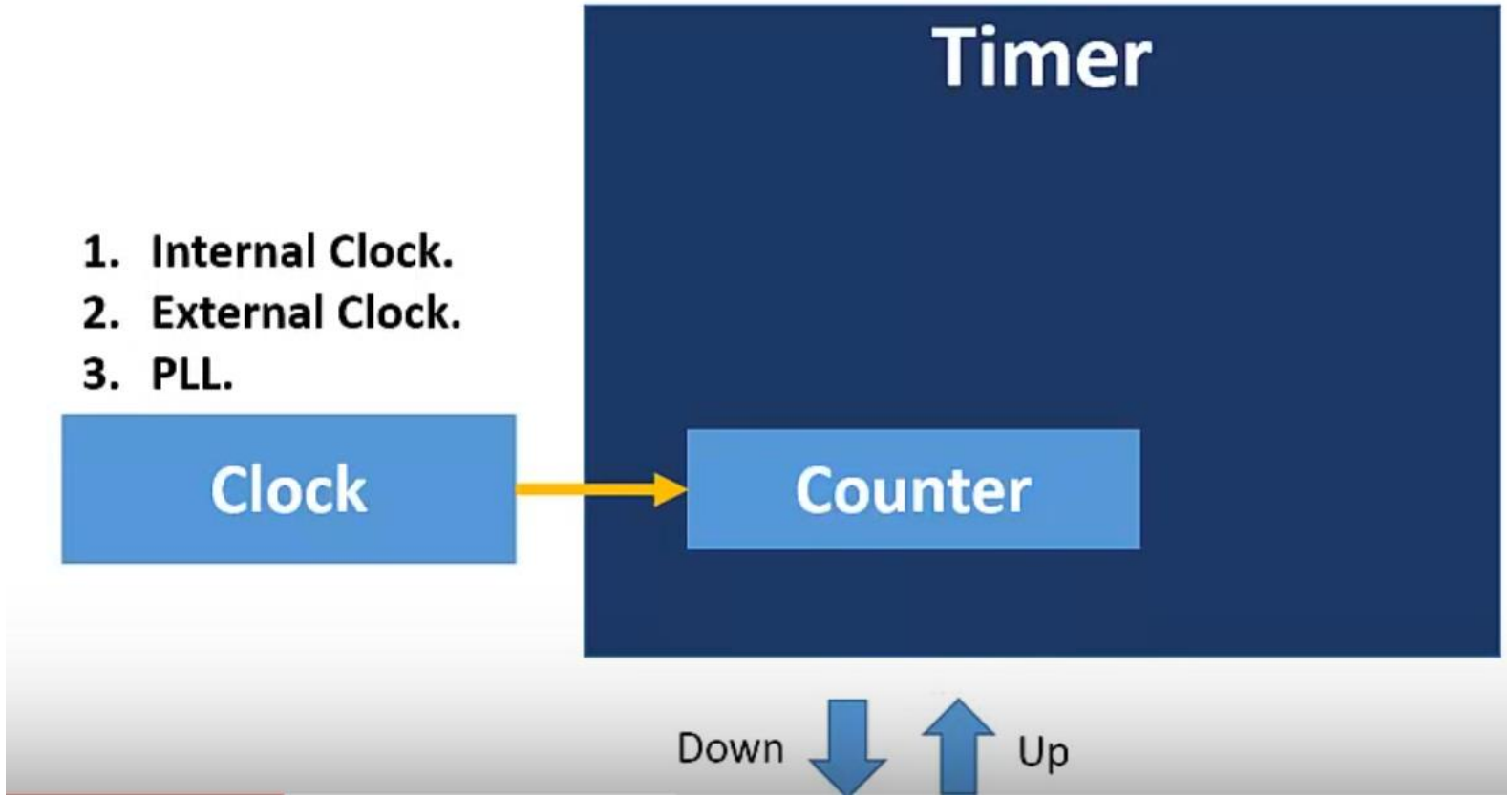


Timer and Counter

- **Timer:**

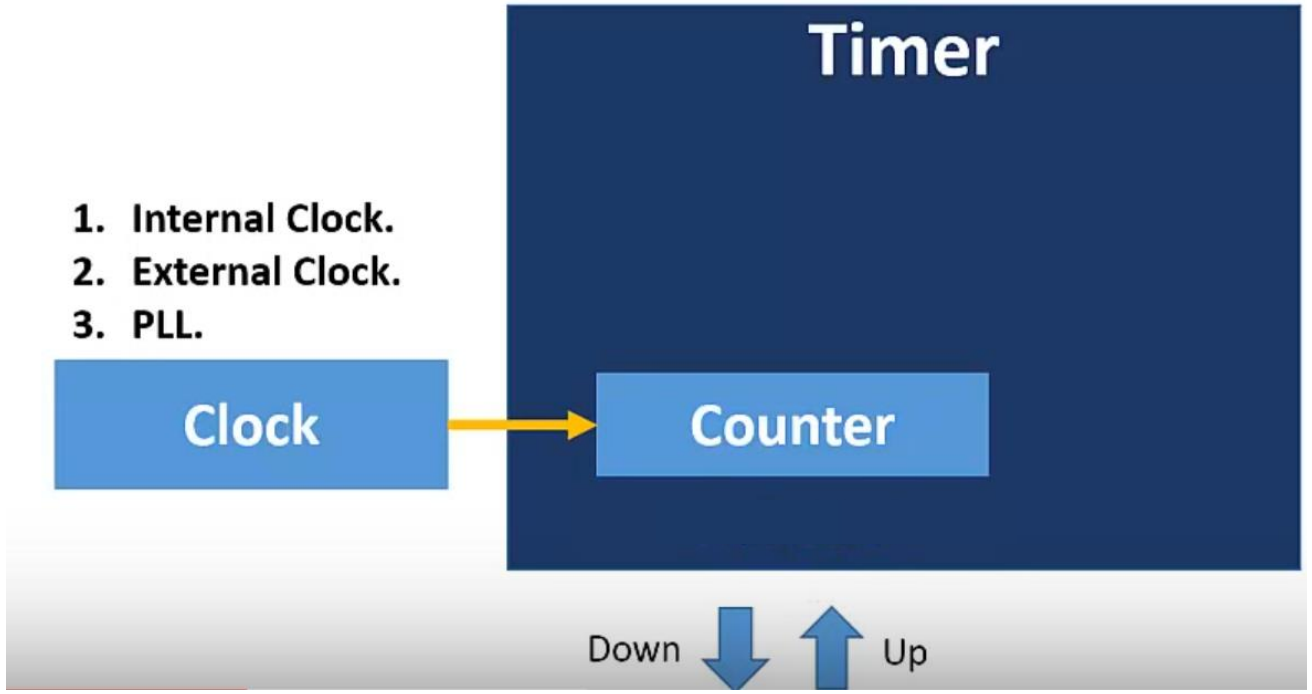
1 Second ---> 16000000 Count

Timer and Counter



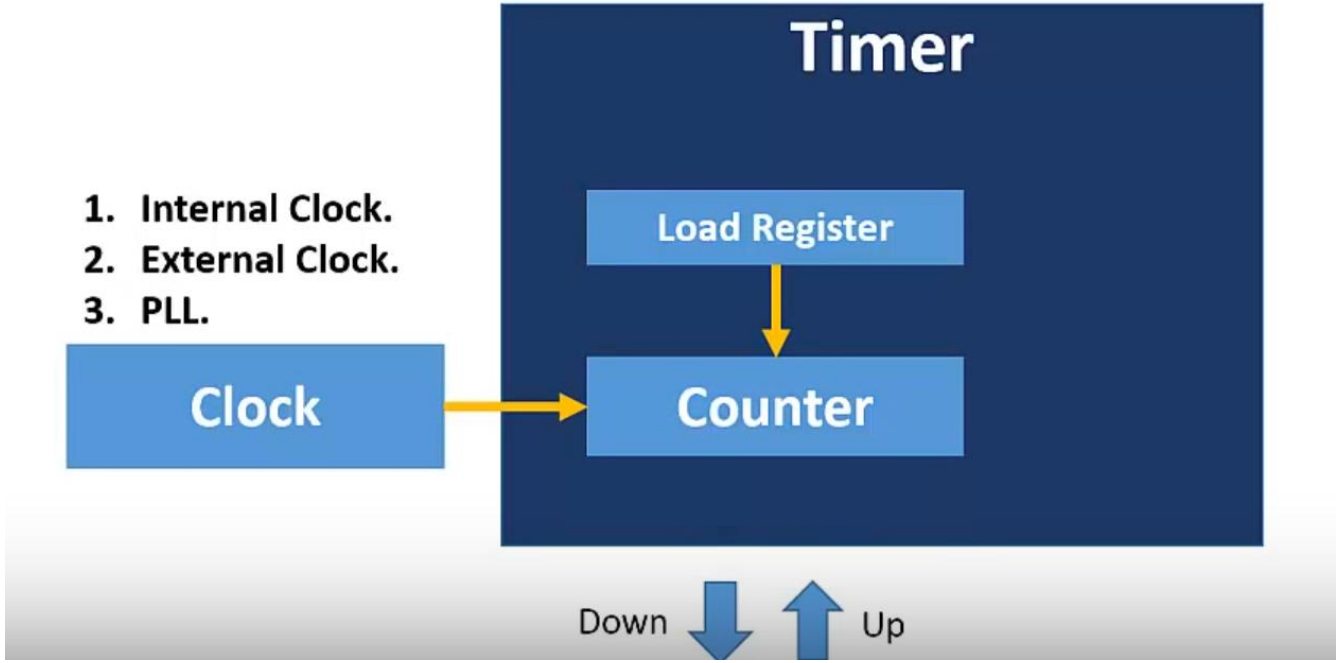
Timer and Counter

0
1
2
.....
65535
65536 Overflow

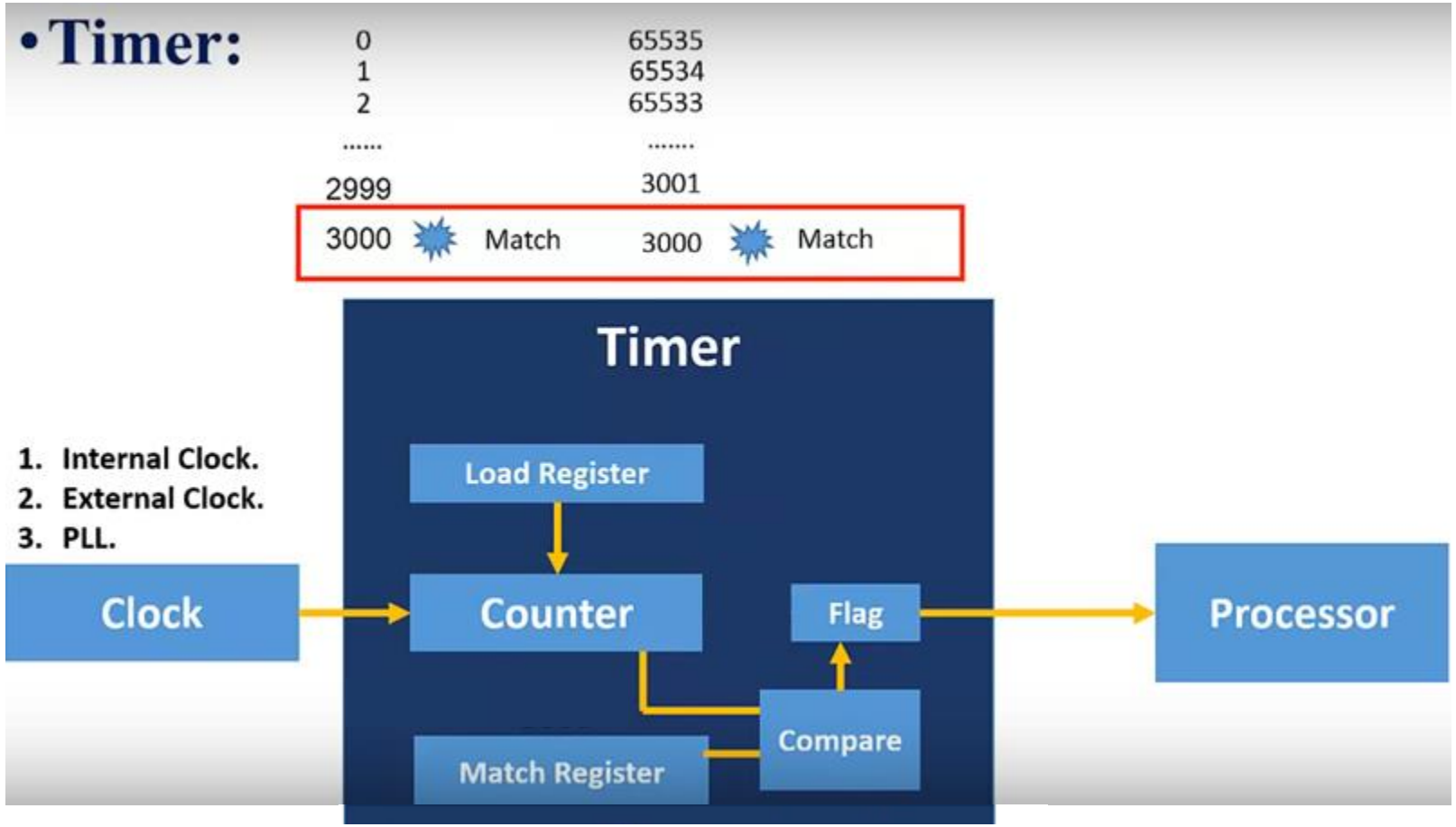


Timer and Counter

0
1
2
.....
65535
65536 ✖ Overflow



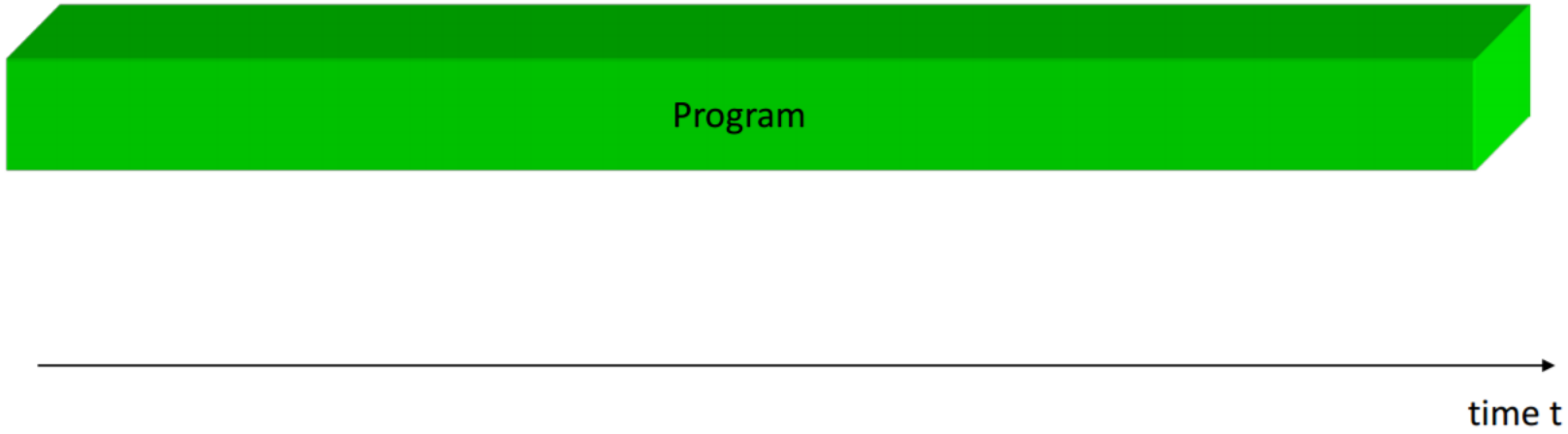
Timer and Counter



Questions

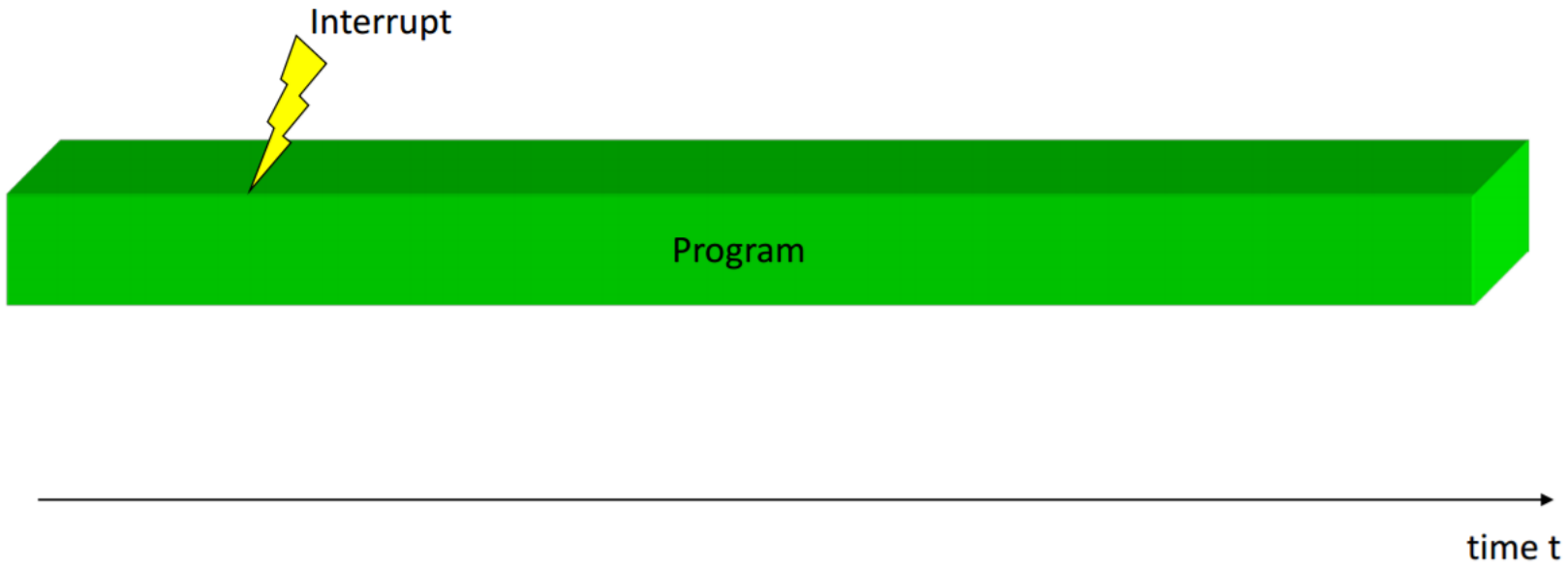
Interrupt Control

Interrupts



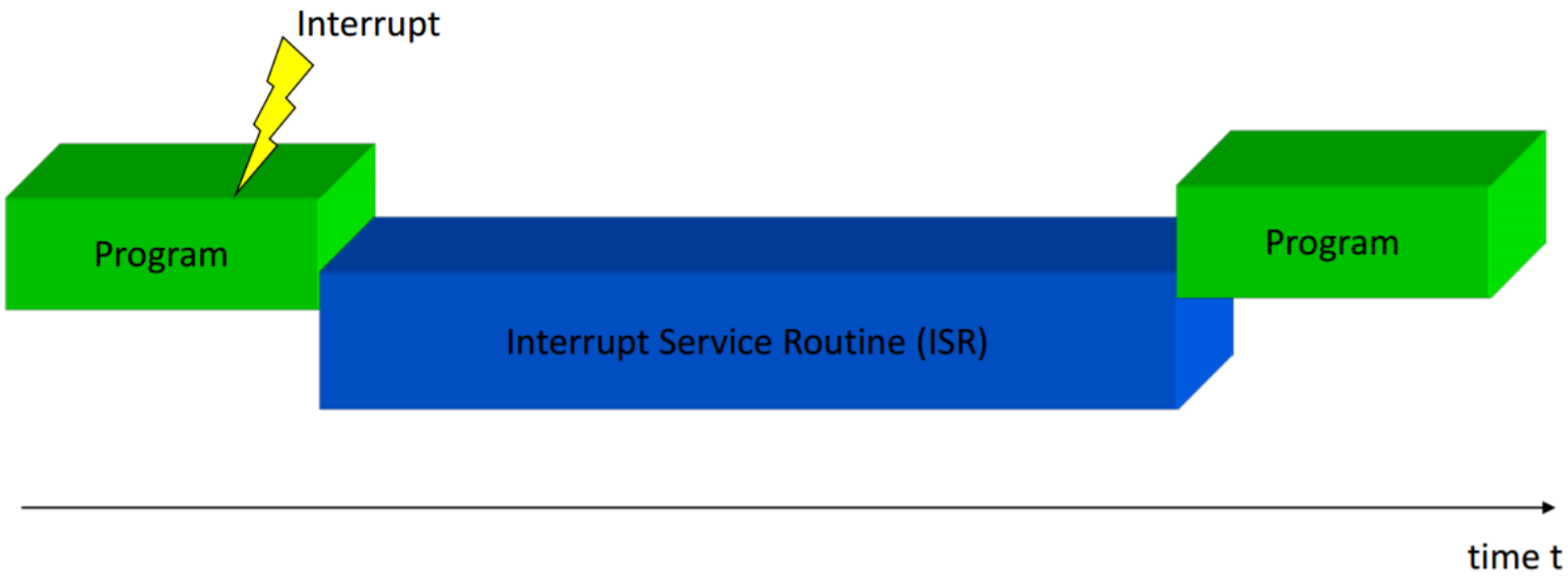
Interrupt Control

Interrupts



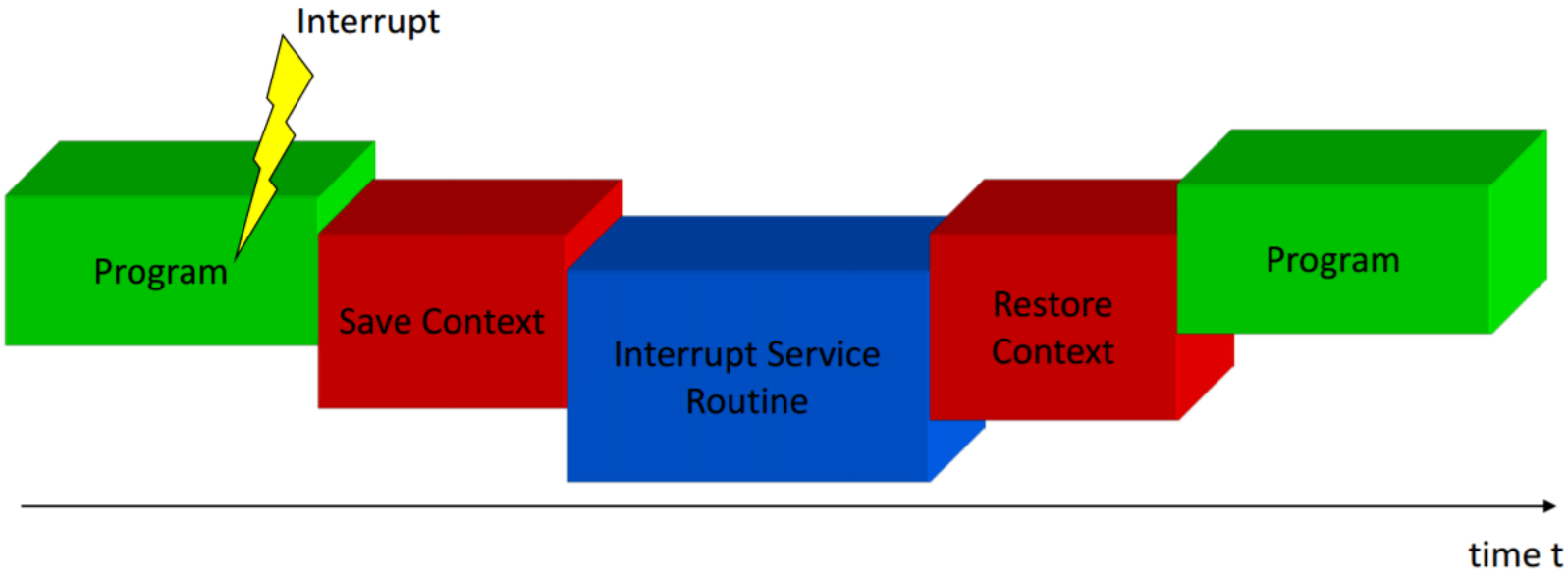
Interrupt Control

Interrupts



Interrupt Control

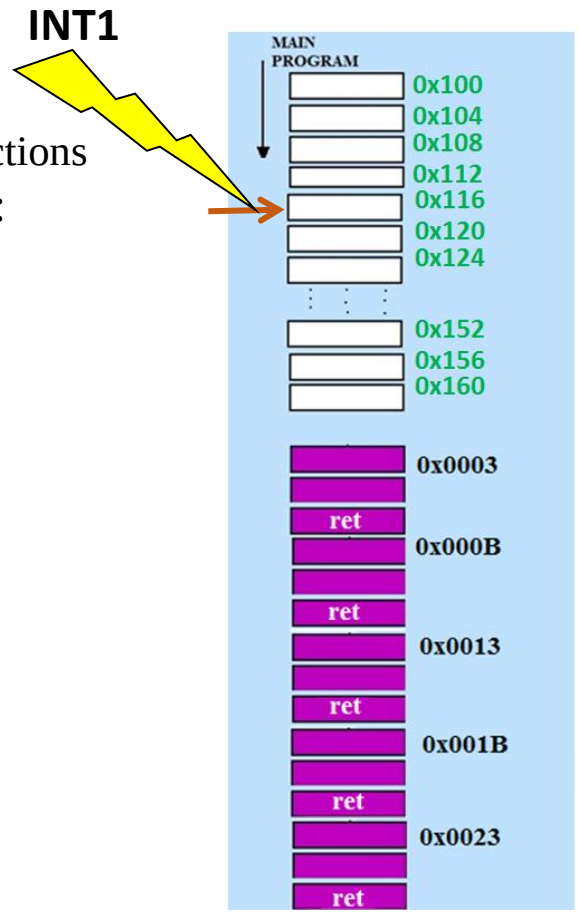
Interrupts



Interrupts

What Happens When an Interrupt Occurs?

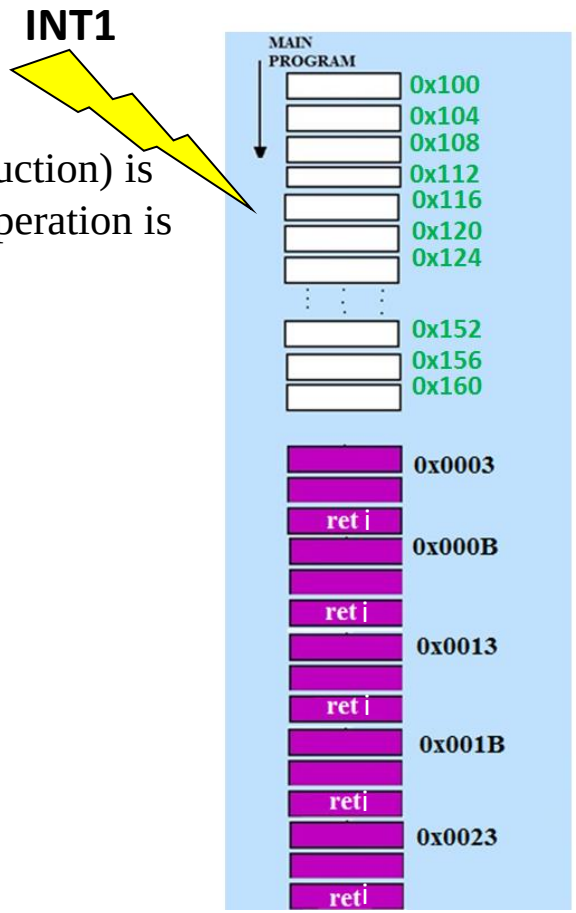
When an interrupt is triggered, the following actions are taken automatically by the microcontroller:

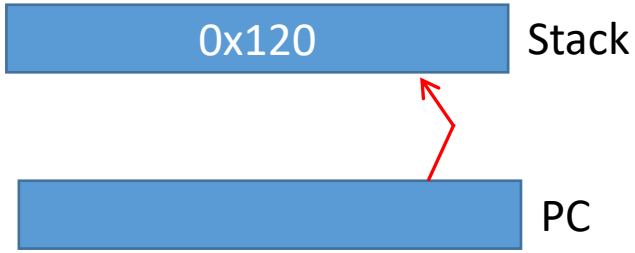


PC (program Counter)

Interrupts

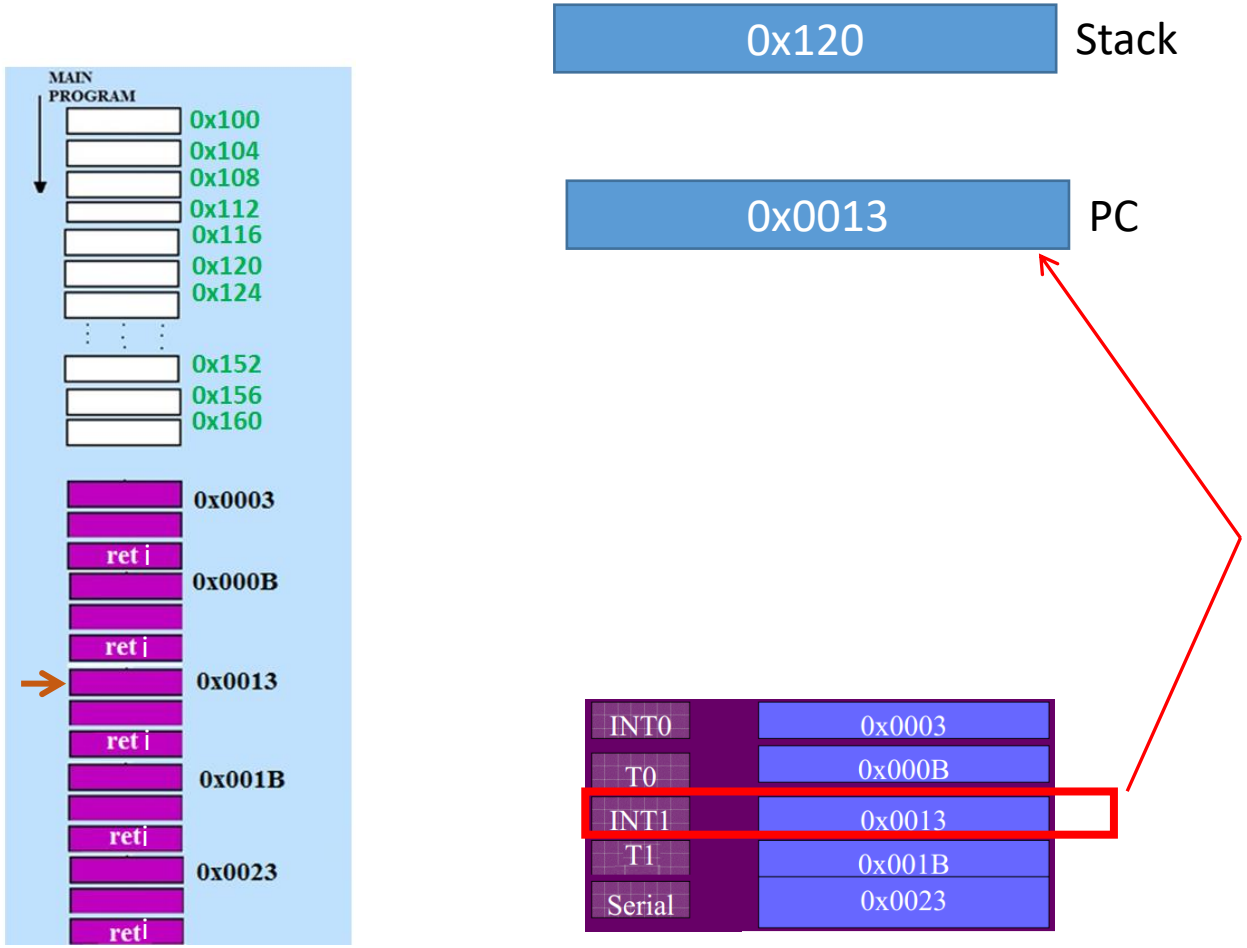
- The value of PC (address of next instruction) is pushed (saved) on stack and normal operation is suspended





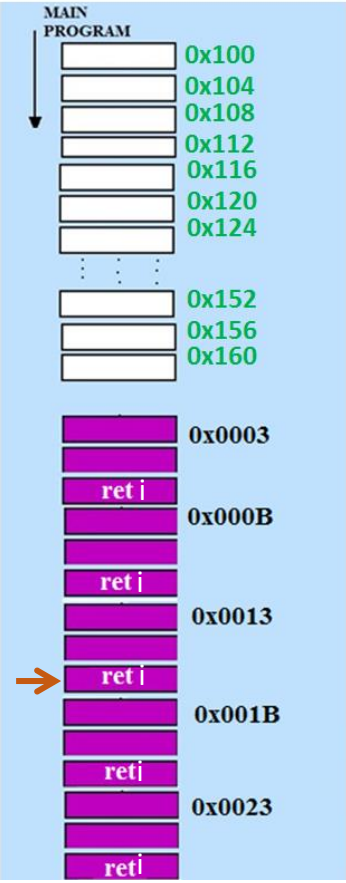
Interrupts

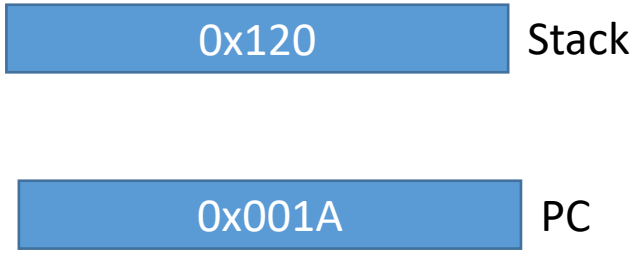
PC is loaded with the address of Interrupt service routine (ISR) from interrupt vector table and ISR is started to execute.



Interrupts

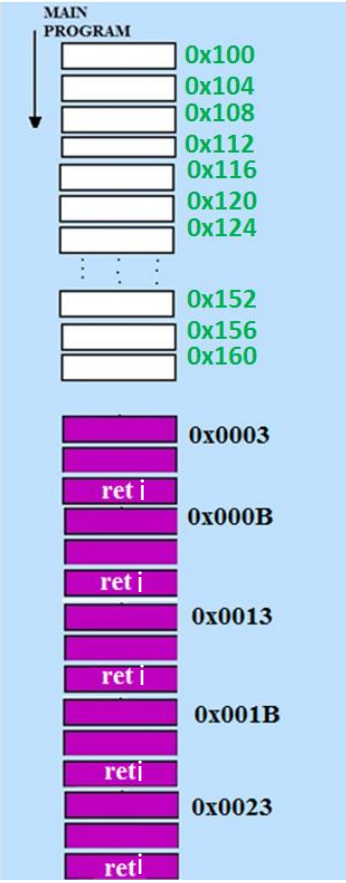
An ISR ends when your program executes the RETI (Return from Interrupt) instruction

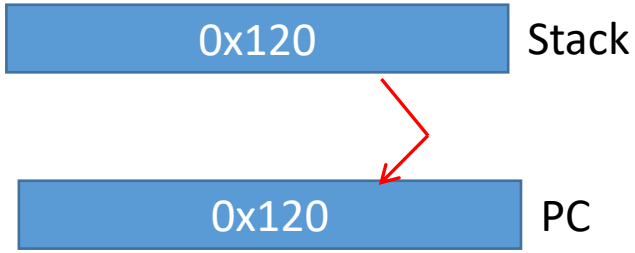




Interrupts

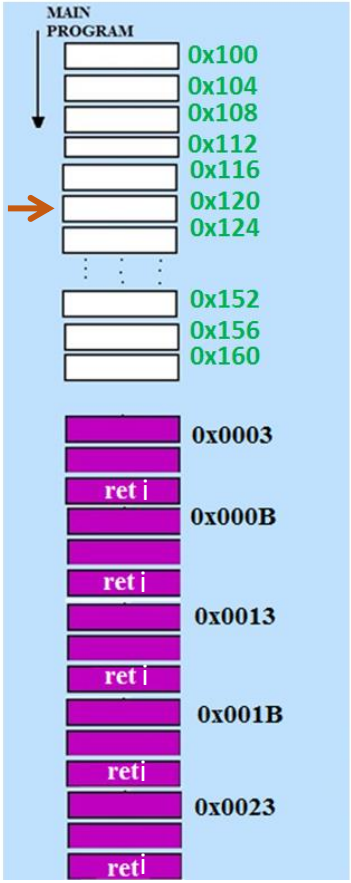
PC is loaded back with the address of the Instruction from stack.





Interrupts

and executions resumes



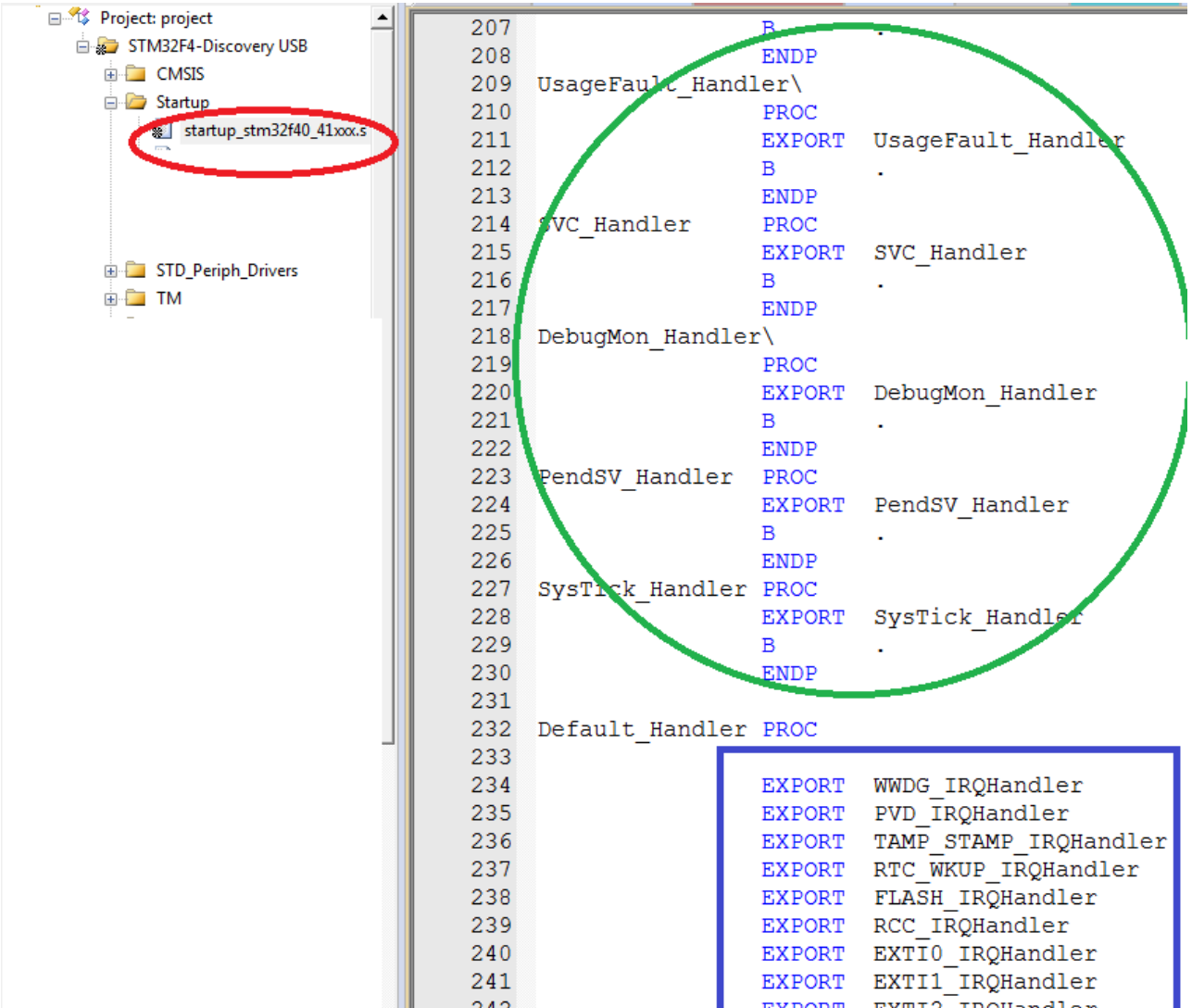
0x120	Stack
0x124A	PC

Interrupt Vector Table

Table 61. Vector table for STM32F405xx/07xx and STM32F415xx/17xx

Position	Priority	Type of priority	Acronym	Description	Address
	-	-	-	Reserved	0x0000 0000
	-3	fixed	Reset	Reset	0x0000 0004
	-2	fixed	NMI	Non maskable interrupt. The RCC Clock Security System (CSS) is linked to the NMI vector.	0x0000 0008
	-1	fixed	HardFault	All class of fault	0x0000 000C
	0	settable	MemManage	Memory management	0x0000 0010
	1	settable	BusFault	Pre-fetch fault, memory access fault	0x0000 0014
	2	settable	UsageFault	Undefined instruction or illegal state	0x0000 0018

How to Execute the Exception / Interrupt Handler




```
                B       .
                ENDP
UsageFault_Handler\
                PROC
                EXPORT  UsageFault_Handler
                B       .
                ENDP
SVC_Handler     PROC
                EXPORT  SVC_Handler
                B       .
                ENDP
DebugMon_Handler\
                PROC
                EXPORT  DebugMon_Handler
                B       .
                ENDP
PendSV_Handler  PROC
                EXPORT  PendSV_Handler
                B       .
                ENDP
SysTick_Handler PROC
                EXPORT  SysTick_Handler
                B       .
                ENDP

Default_Handler PROC

                EXPORT  WWDG_IRQHandler
                EXPORT  PVD_IRQHandler
                EXPORT  TAMP_STAMP_IRQHandler
                EXPORT  RTC_WKUP_IRQHandler
                EXPORT  FLASH_IRQHandler
                EXPORT  RCC_IRQHandler
                EXPORT  EXTI0_IRQHandler
                EXPORT  EXTI1_IRQHandler
```

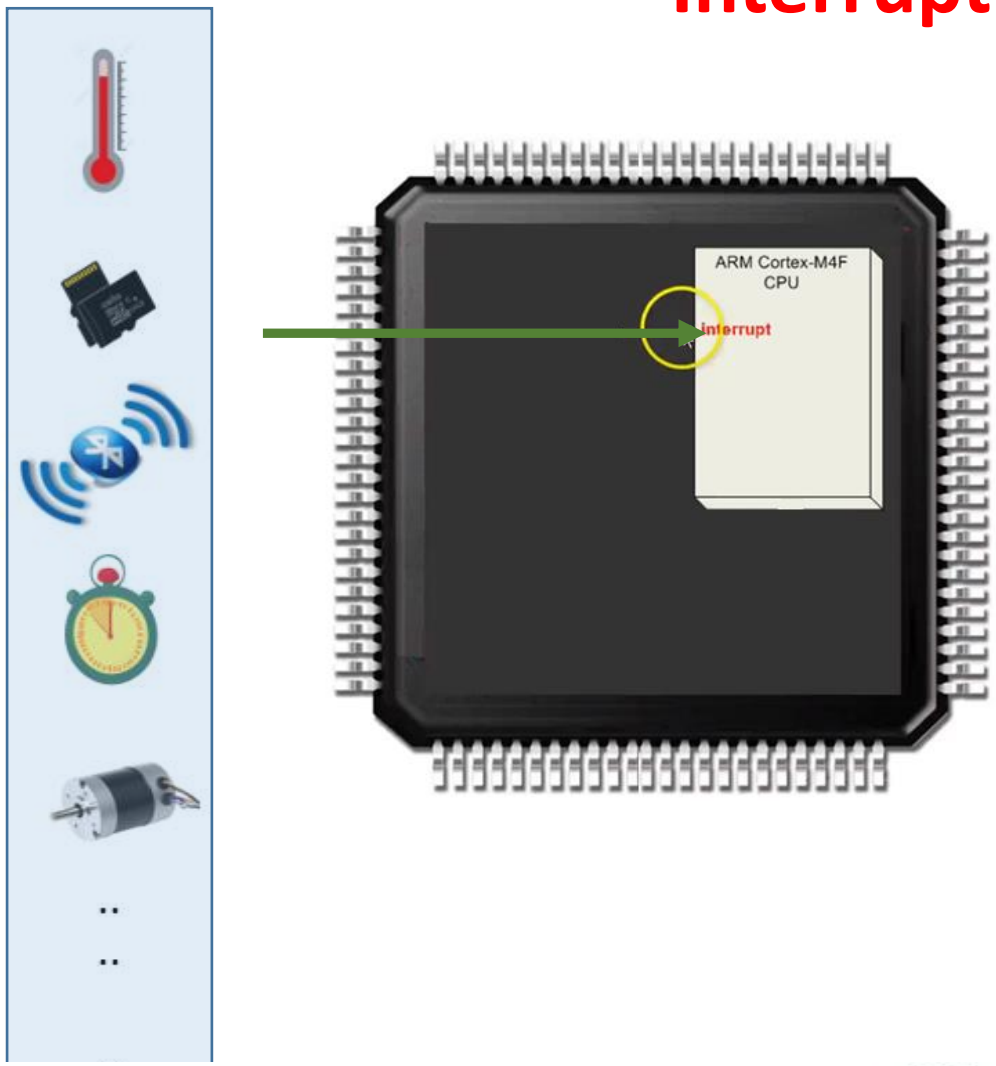
Nested Vector Interrupt Controller (NVIC)

```
NVIC(Nested Vector interrupt Controller) discussion
============================

1) NVIC is one of the peripheral of the Cortex M processor core.

2) it is used to configure the 240 interrupts

3) using NVIC registers you can Enable/Disable/Pend various interrupts and
read the status of the active and pending interrupts .

4) you can configure the priority and priority grouping of various interrupts.

5)Its is called as "Nested" because, it supports pre-empting a lower priority
interrupt handler when higher priority interrupt arrives.
```

Nested Vector Interrupt Controller (NVIC)

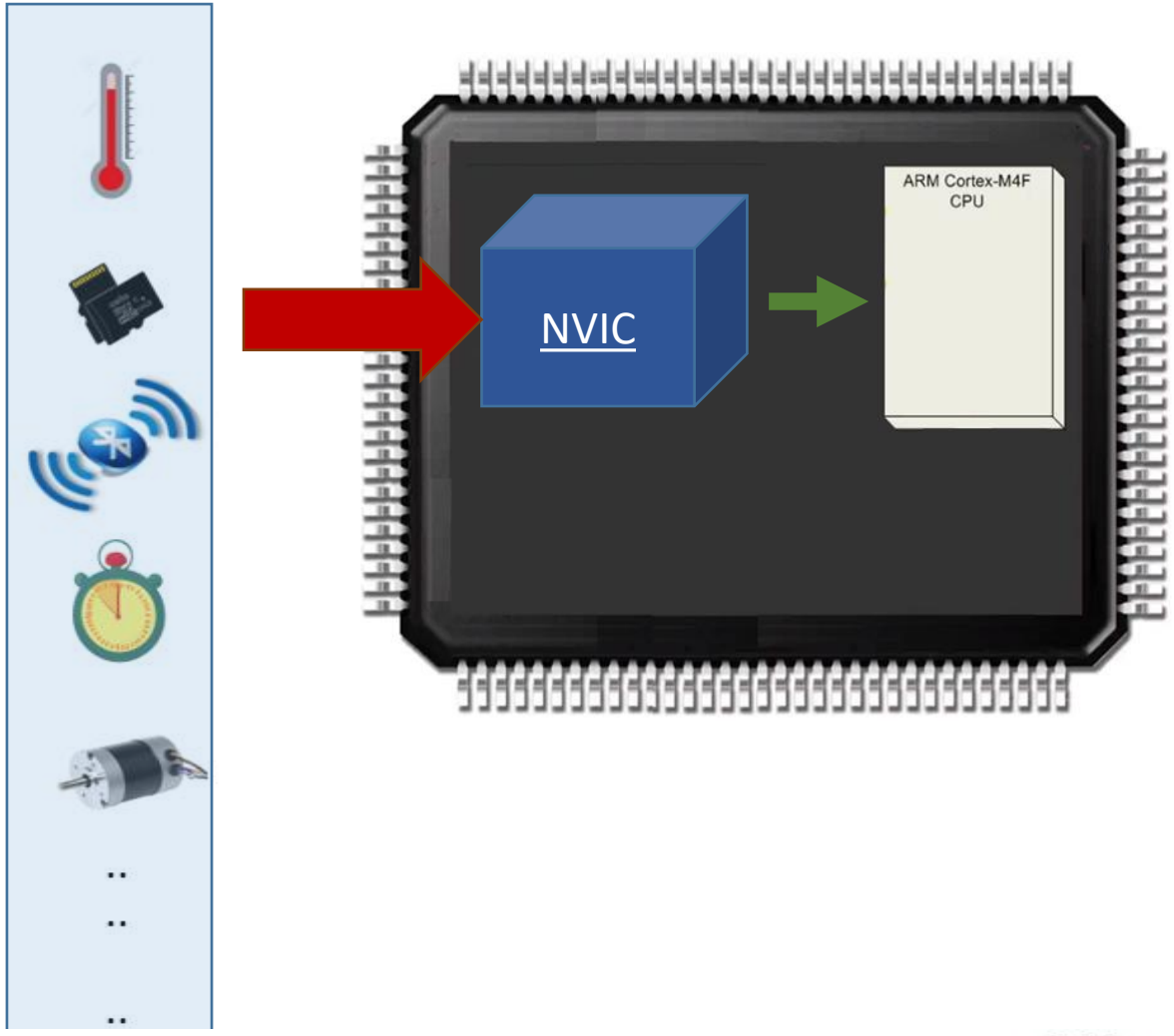
Interrupt

- In the old days, a CPU would have one interrupt lines.
- In this case, either an external peripheral, or just software checking lots of status registers would be used to determine the cause of the interrupt, and look up the location of the relevant handler.



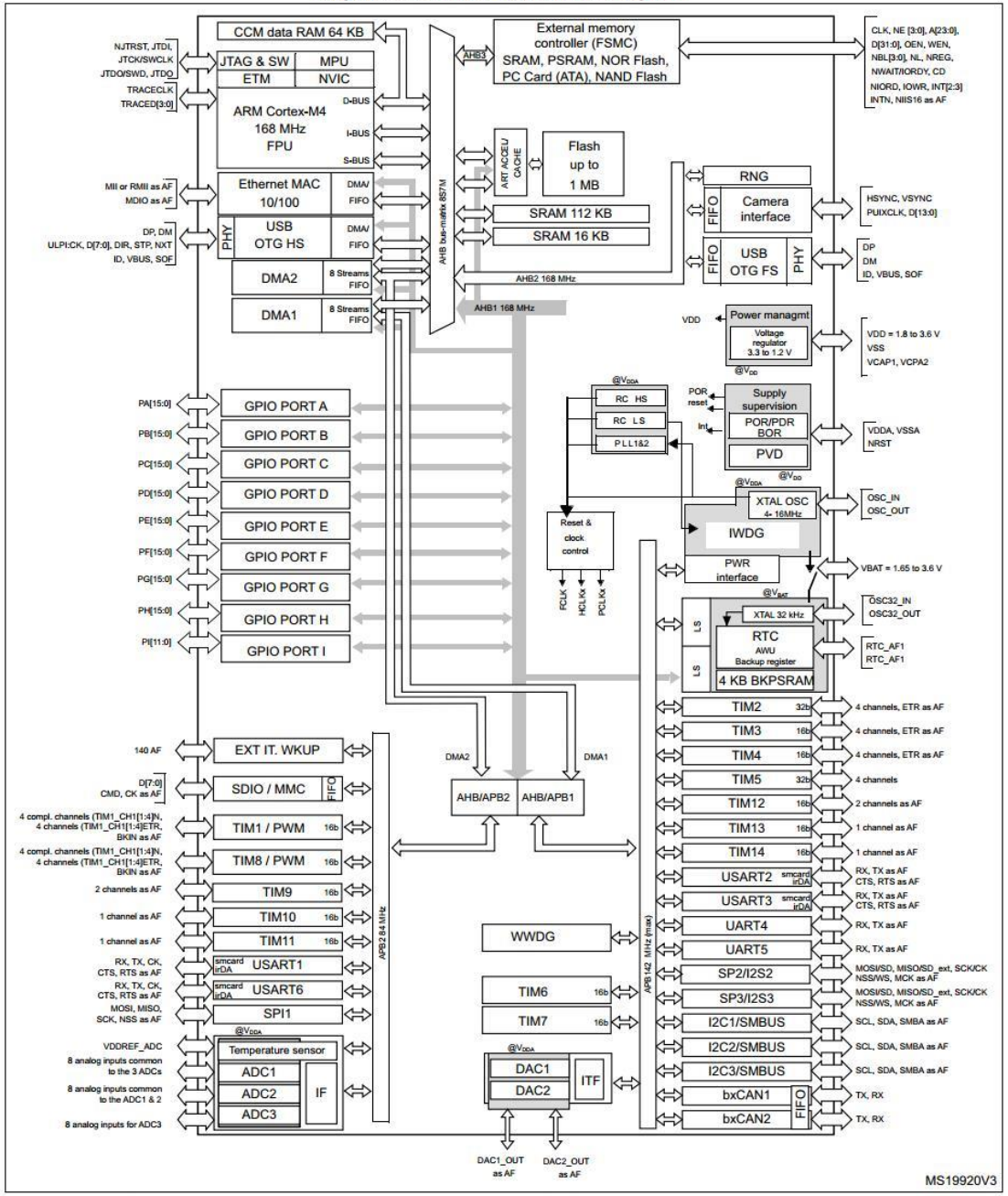
Nested Vector Interrupt Controller (NVIC)

Particularly in a real time system, time to get to the first real instruction of an interrupt handler is important,
so **V**ectored **I**nterrupt **C**ontrollers were introduced to provide the right address straight off to the processor, saving a few instructions.



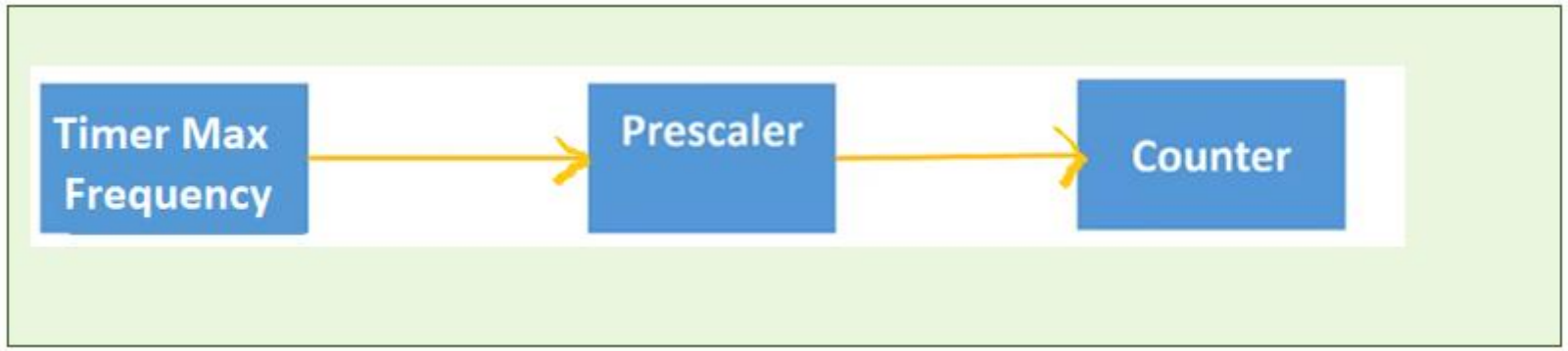
Exercise

Figure 5. STM32F40x block diagram



1. The timers connected to APB2 are clocked from TIMxCLK up to 168 MHz, while the timers connected to APB1 are clocked from TIMxCLK either up to 84 MHz or 168 MHz, depending on TIMPRE bit configuration in the RCC_DCKCFGR register.
2. The camera interface and ethernet are available only on STM32F407xx devices.

- **Timer:**
  - **Solution 3 (Use Prescaler):**



Timer type	Timer	Counter Resolution	Counter type	Prescaler Factor	DMA Request Generation	Capture/ Compare Channels	Complementary Output	Interface Clock (TIMx_CK_INT)	Max Timer Clock (MHz)
Advanced Control	TIM1, TIM8	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	Yes	APB2 Timer Clock	168
General Purpose	TIM2, TIM5	32-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	No	APB1 Timer Clock	84
	TIM3, TIM4	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	No	APB1 Timer Clock	84
	TIM9	16-bit	Up	between 1 and 65536	No	2	No	APB2 Timer Clock	168
	TIM10, TIM11	16-bit	Up	between 1 and 65536	No	1	No	APB2 Timer Clock	168
	TIM12	16-bit	Up	between 1 and 65536	No	2	No	APB1 Timer Clock	84
	TIM13, TIM14	16-bit	Up	between 1 and 65536	No	1	No	APB1 Timer Clock	84
Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

Timer type	Timer	Counter Resolution	Counter type	Prescaler Factor	DMA Request Generation	Capture/ Compare Channels	Complementary Output	Interface Clock (TIMx_CK_INT)	Max Timer Clock (MHz)
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	TIM10, TIM11	16-bit	Up	between 1 and 65536	No	1	No	APB2 Timer Clock	168
	TIM12	16-bit	Up	between 1 and 65536	No	2	No	APB1 Timer Clock	84
	TIM13, TIM14	16-bit	Up	between 1 and 65536	No	1	No	APB1 Timer Clock	84
Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

- **TIMx Counter** (TIMx_CNT, address offset 0x24)
  - 16-bit binary counter (32 in TIM2, TIM5)

Timer type	Timer	Counter Resolution	Counter type	Prescaler Factor	DMA Request Generation	Capture/ Compare Channels	Complementary Output	Interface Clock (TIMx_CK_INT)	Max Timer Clock (MHz)
Advanced Control	TIM1, TIM8	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	Yes	APB2 Timer Clock	168
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	TIM13, TIM14	16-bit	Up	between 1 and 65536	No	1	No	APB1 Timer Clock	84
	TIM6,			between 1				APB1 Timer	

Timer type	Timer	Counter Resolution	Counter type	Prescaler Factor	DMA Request Generation	Capture/ Compare Channels	Complementary Output	Interface Clock (TIMx_CK_INT)	Max Timer Clock (MHz)
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Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

- **TIMx Prescale Register** (TIMx_PSC, address offset 0x28)
  - Clock prescale value (16 bits)

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  - Clock prescale value (16 bits)

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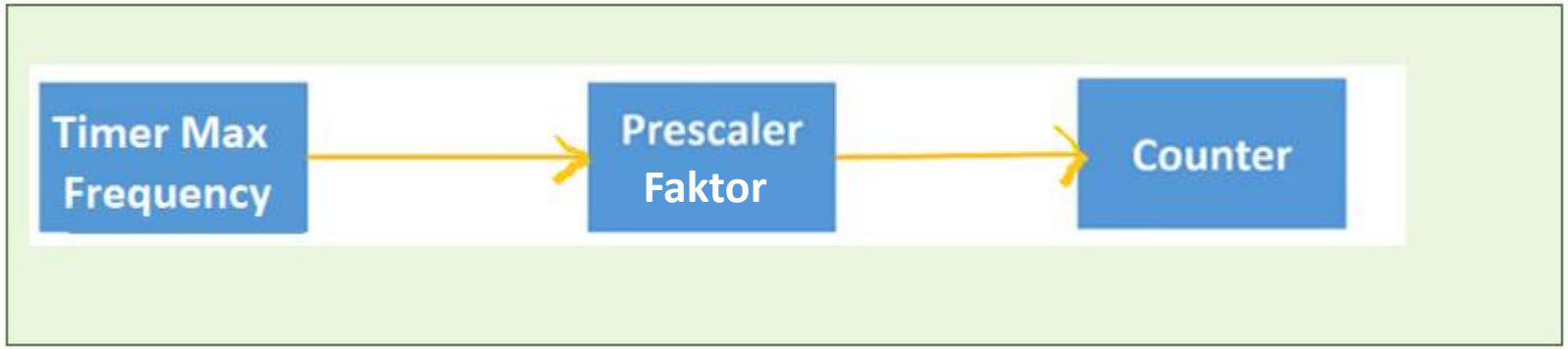
Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

$2^{16} \rightarrow$ 65,536

The prescaler is always a 16-bit integer value in the range 0 .. 65535, representing a division by 1 .. 65536

The prescaler is always a 16-bit integer value in the range 0 .. 65535
Representing a division by 1 .. 65536



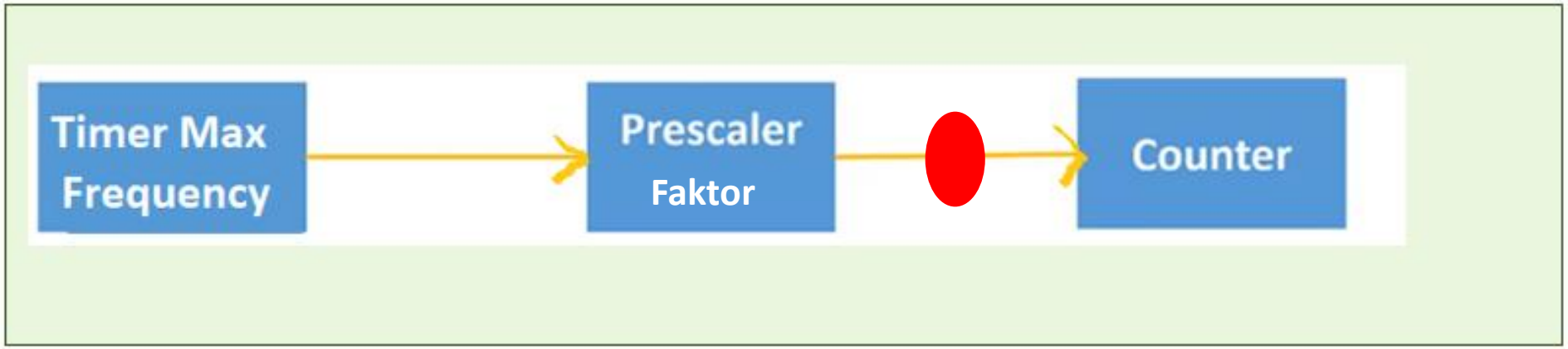
Prescalar Factor = 2^x

$x \longrightarrow$ Prescalar Value

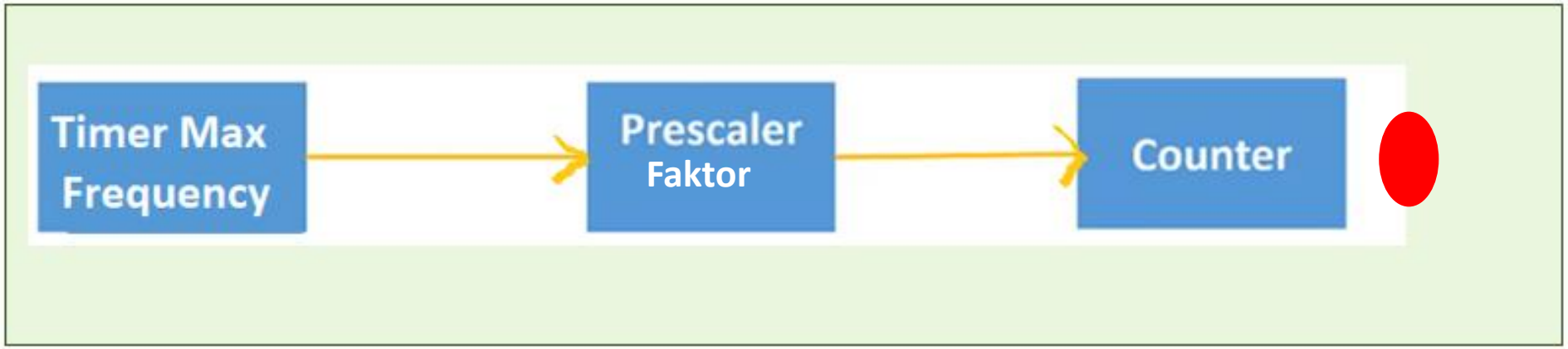
```
TIM_TimeBaseStructure.TIM_Prescaler =
```

- Modify the blinking frequency in Timer_Template.cpp file to 0.5Hz

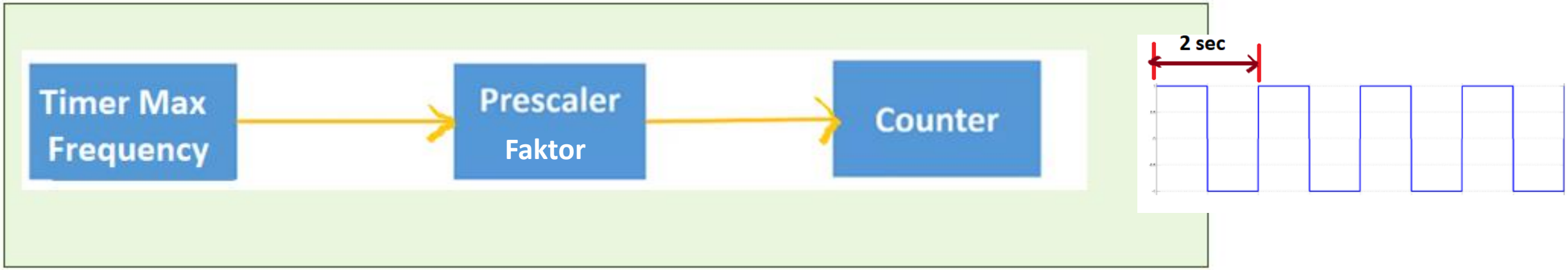
- Modify the blinking frequency in Timer_Template.cpp file to 0.5Hz



- Modify the blinking frequency in Timer_Template.cpp file to 0.5Hz



- Modify the blinking frequency in Timer_Template.cpp file to 0.5Hz



- Modify the blinking frequency in Timer_Template.cpp file to 0.5Hz

$$\boxed{Prescaler = \frac{Timer_Max_Frequency}{Frequency_After_Prescaler} - 1}$$

Value

$$\boxed{Period = \frac{Frequency_After_Prescaler}{Timer_Desired_Frequency} - 1}$$

- Modify the blinking frequency in Timer_Template.cpp file to 0.5Hz

$$Prescaler = \frac{Timer_Max_Frequency}{Frequency_After_Prescaler} - 1$$

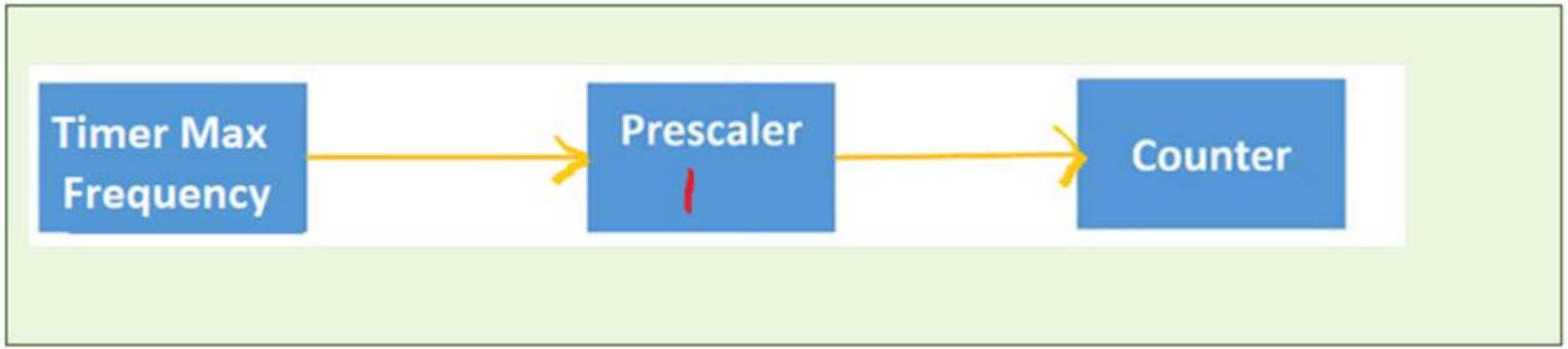
$$\frac{84\ MHz}{84\ MHz} - 1 = 0$$

So

- Prescalar Value = 0

  Prescalar Factor = 1

$$Period = \frac{Frequency_After_Prescaler}{Timer_Desired_Frequency} - 1$$



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Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

$$\boxed{Prescaler = \frac{Timer_Max_Frequency}{Frequency_After_Prescaler} - 1}$$

$$\boxed{Period = \frac{Frequency_After_Prescaler}{Timer_Desired_Frequency} - 1}$$

$$\text{Period} = \frac{84\ \text{MHz}}{0.5\ \text{Hz}} - 1$$

Period = 167 999 999

Timer type	Timer	Counter Resolution	Counter type	Prescaler Factor	DMA Request Generation	Capture/ Compare Channels	Complementary Output	Interface Clock (TIMx_CK_INT)	Max Timer Clock (MHz)
Advanced Control	TIM1, TIM8	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	Yes	APB2 Timer Clock	168
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Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

The Counter can count from 0 to **65535**

Timer type	Timer	Counter Resolution	Counter type	Prescaler Factor	DMA Request Generation	Capture/ Compare Channels	Complementary Output	Interface Clock (TIMx_CK_INT)	Max Timer Clock (MHz)
Advanced Control	TIM1, TIM8	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	Yes	APB2 Timer Clock	168
General Purpose	TIM2, TIM5	32-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	No	APB1 Timer Clock	84
	TIM3, TIM4	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	No	APB1 Timer Clock	84
	TIM9	16-bit	Up	between 1 and 65536	No	2	No	APB2 Timer Clock	168
	TIM10, TIM11	16-bit	Up	between 1 and 65536	No	1	No	APB2 Timer Clock	168
	TIM12	16-bit	Up	between 1 and 65536	No	2	No	APB1 Timer Clock	84
	TIM13, TIM14	16-bit	Up	between 1 and 65536	No	1	No	APB1 Timer Clock	84
Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

The Counter can count from 0 to 65535

Period = 167 999 999

Timer type	Timer	Counter Resolution	Counter type	Prescaler Factor	DMA Request Generation	Capture/ Compare Channels	Complementary Output	Interface Clock (TIMx_CK_INT)	Max Timer Clock (MHz)
Advanced Control	TIM1, TIM8	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	Yes	APB2 Timer Clock	168
General Purpose	TIM2, TIM5	32-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	No	APB1 Timer Clock	84
	TIM3, TIM4	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	No	APB1 Timer Clock	84
	TIM9	16-bit	Up	between 1 and 65536	No	2	No	APB2 Timer Clock	168
	TIM10, TIM11	16-bit	Up	between 1 and 65536	No	1	No	APB2 Timer Clock	168
	TIM12	16-bit	Up	between 1 and 65536	No	2	No	APB1 Timer Clock	84
	TIM13, TIM14	16-bit	Up	between 1 and 65536	No	1	No	APB1 Timer Clock	84
Basic	TIM6, TIM7	16-bit	Up	between 1 and 65536	Yes	0	No	APB1 Timer Clock	84

Period = 167 999 999

The Counter can count upto

2 ^ 32 =

4,294,967,296

```
void Setup_Timer(void)
{
  /* TIM2 clock enable */
  RCC_APB1PeriphClockCmd(RCC_APB1Periph_TIM2, ENABLE);
            .
            .
            .
            .
            .
            .
            .

  TIM_TimeBaseInit(TIM2, &TIM_TimeBaseStructure);
  /* TIM2 enable counter */
  TIM_Cmd(TIM2, ENABLE);


}
```